

QuickStart to Research Data Management

Team RDM Support

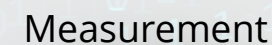
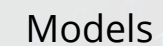
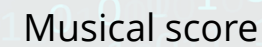
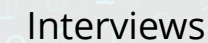
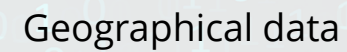
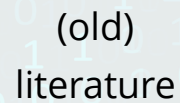
Today

Key concepts

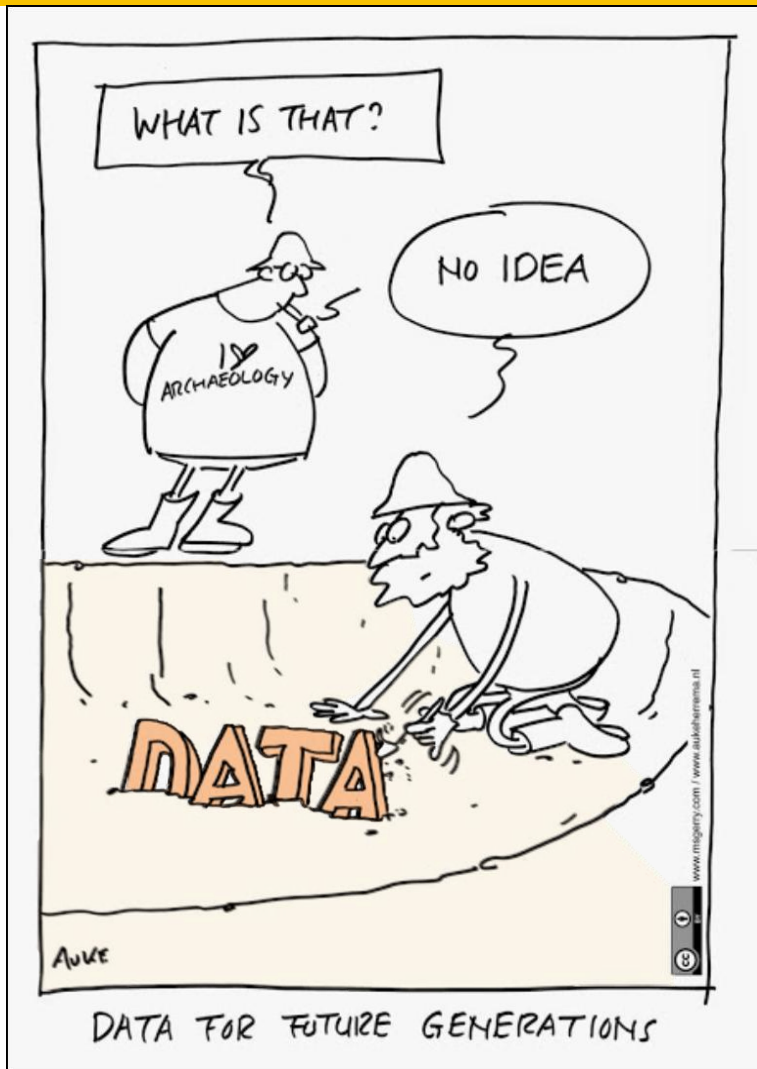
- Data
- Research Data Lifecycle
- Research Data Management
- FAIR data
- Open data

Data management throughout the research data lifecycle

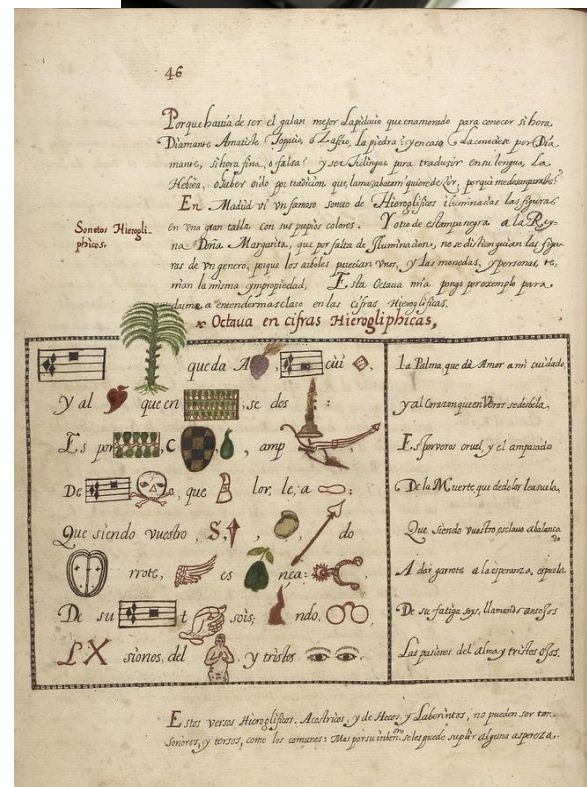
- Data Management Plans (DMP)
- Data collection
- Organizing your data
- Data documentation
- Storage
- Archiving
- Publishing data
- Data re-use



Making your data last



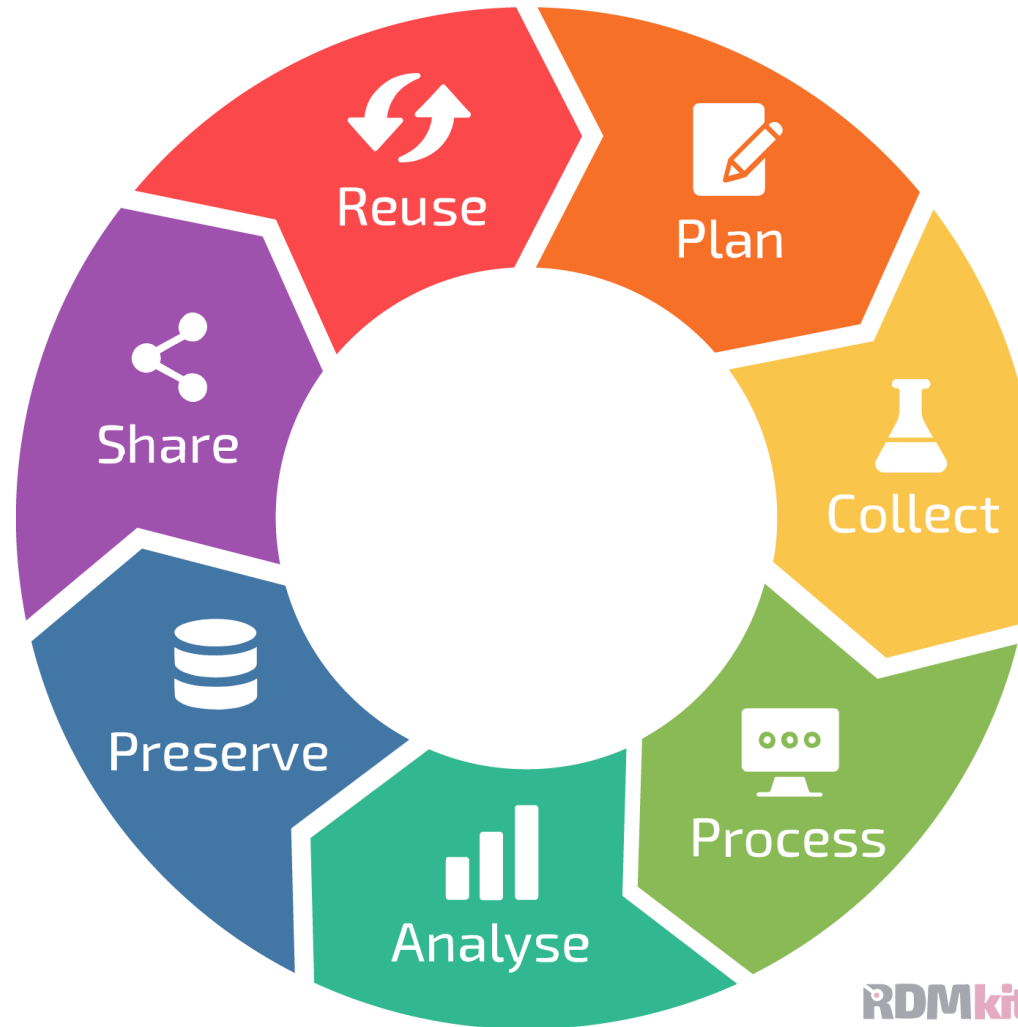
Auke Herrema - Het Bouwteam (2014)



Name	Date modified	Type
7a1afd984e29b9f1190684	8/28/2016 8:31 PM	File folder
8b7aa20d48de418801436518	8/11/2016 3:11 AM	File folder
82deebd23d8251b2986245	8/11/2016 12:52 PM	File folder
387c52c1af7496d51b	8/13/2016 7:41 PM	File folder
453f3e71eb2683b43ffa351fc03a	8/10/2016 5:41 PM	File folder
485d4dd6fd8e3c6e16	8/23/2016 4:30 PM	File folder
753ba357f17fc7c04953eb8545cf	10/3/2014 2:06 PM	File folder
9571af284a659d81530b1183e625	8/12/2016 9:40 PM	File folder
a2c3980f5985765470ea32	8/11/2016 8:10 PM	File folder
a08495c1ecf8be8aa3	8/13/2016 5:59 PM	File folder
Asus A55VM	7/28/2016 12:55 PM	File folder
be1c208b5d22d402c3dcf18d26cf99	10/2/2014 3:00 AM	File folder
be614ee0171cf84a51b22b46a2c607	8/28/2016 6:33 PM	File folder
bf64731fc981be2f7a867c62f8b32b9d	8/29/2016 6:00 PM	File folder
df4ff9cdd1c9f2da9495d508243b9f56	8/15/2016 12:54 AM	File folder
e4f19a4a3a440ac8b216ba0620	8/23/2016 8:12 PM	File folder

Here's my data... good luck!

The Research Data Lifecycle

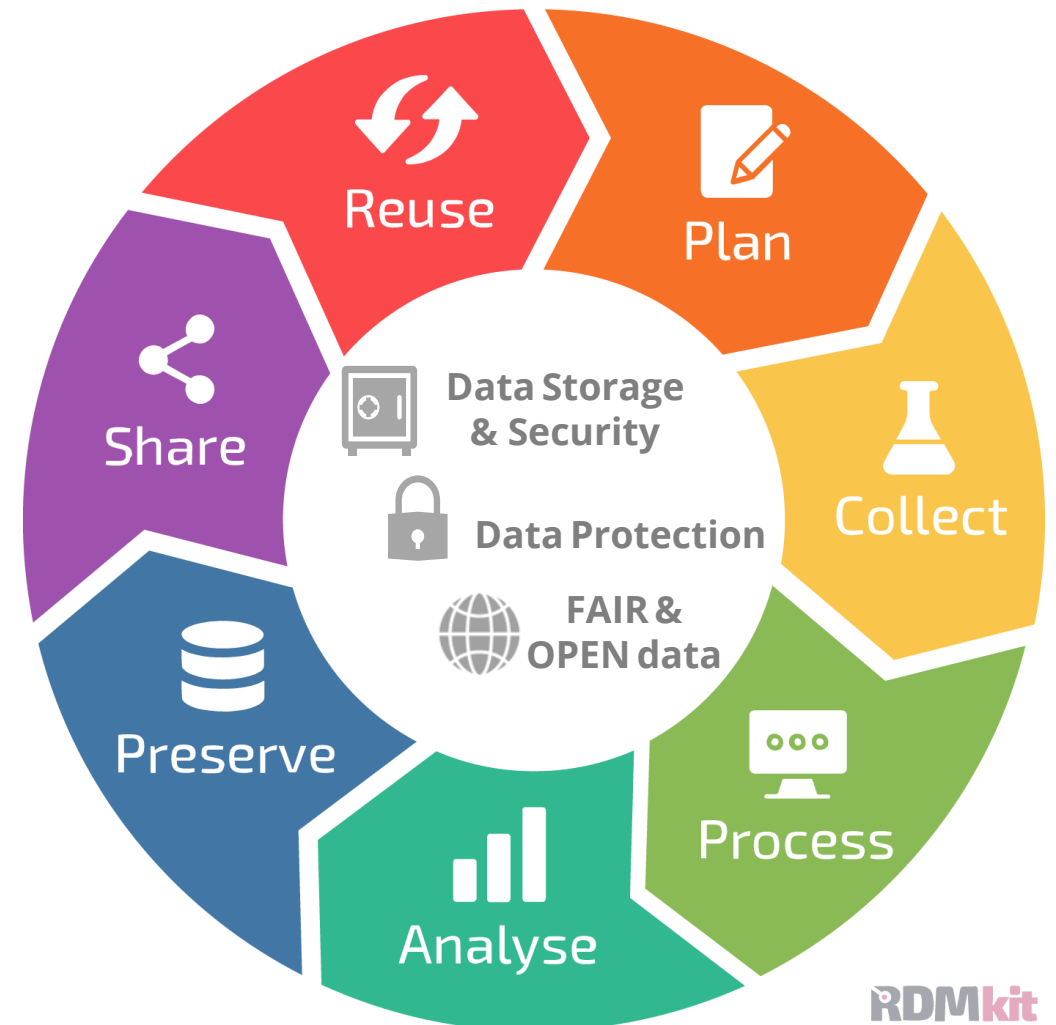


Research Data Management (RDM)

RDM refers to the **active organization and maintenance of data** created during a research project.

It is an **ongoing activity** **throughout the data lifecycle**, from initial planning to suitable archiving of the data at the project's completion.

RDM should be guided by the FAIR principles

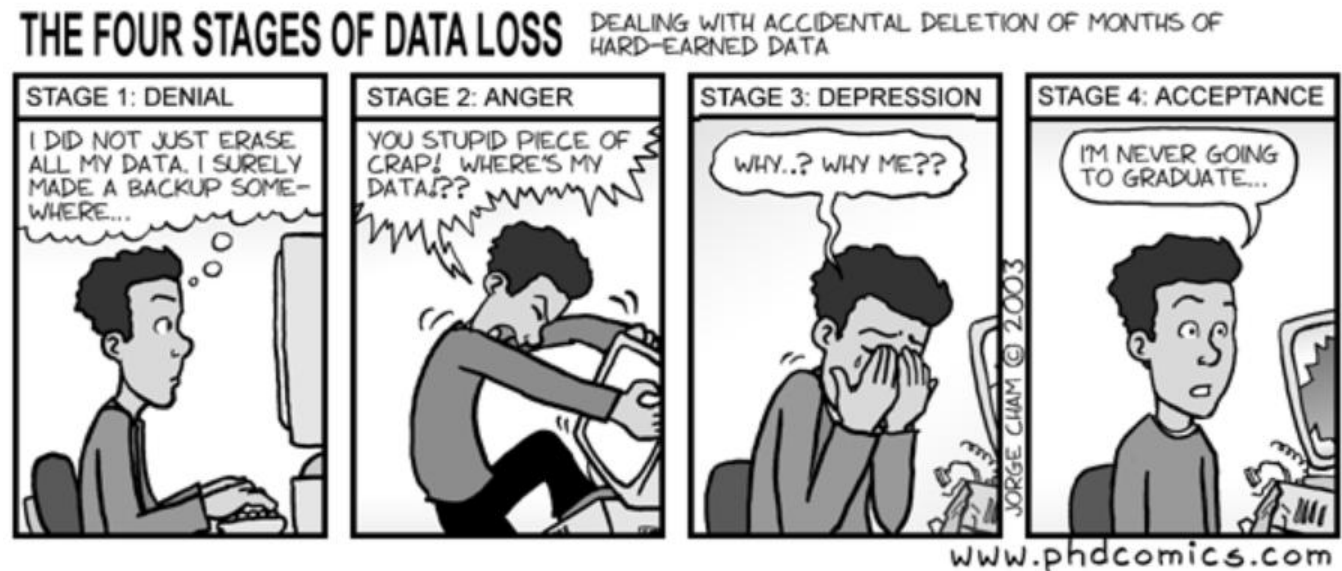


Importance of RDM

Planning in advance and investing in Good (FAIR) DM Practices



- Easy (long term) **data access** for yourself & colleagues
- Avoiding **data loss (real or virtual)**
- Avoiding **data redundancy**
- Streamline documentation and project structure
- Protecting data (Intellectual Property Rights)
- ...



Good Scientific Practice

Proper RDM is crucial for:

- **Reproducibility**

Data can be reused by yourself or other research groups

- **Research integrity**

Verification & transparency

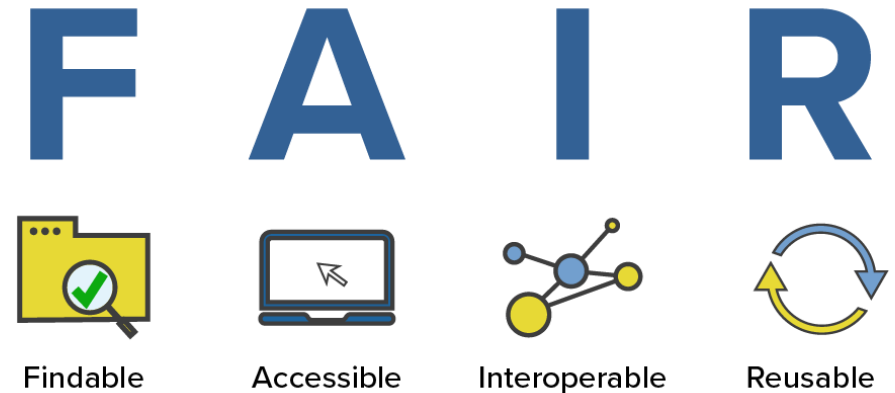
- **Funder mandates and policies**

NWO, ERC, Horizon, ZonMW require DMP for every project

- **Legislation**

Personal data must abide by GDPR regulations

Read more about research integrity at the UU on our website [Research integrity](#)



*“as **open** as possible, as closed as necessary...
But **always FAIR**”*

The FAIR data principles

FINDABILITY

Possibility for humans and machines to **discover** your data
(*metadata, Persistent Identifiers*)

F

A

I

R



Findable



Accessible



Interoperable



Reusable



INTEROPERABILITY

Data and metadata conform to **recognized formats and standards** which allows them to be combined and exchanged



REUSABILITY

Lots of **documentation**, which is needed to support data and **interpretation** and reuse

ACCESSIBILITY

Possibility for humans and machines to **gain access** to your data
(*specific conditions or restrictions where appropriate*)



Open Data

“ Open Data is data that can be **freely used, re-used and redistributed by anyone** subject only, at most, to the requirements to attribute and share-alike ”

([The Open Data Handbook](#))

Reasons data can be FAIR yet not open:

- ☐ Ethical or cultural obligations
- ☐ Safety and security risks
- ☐ Contractual/licensing constraints
- ☐ National security/export controls
- ☐ Protection of sensitive ecological or heritage sites
- ☐ Embargoes for ongoing research
- ☐ Economic or political sensitivity
- ☐ Risks to vulnerable communities



F A I R

Make your data...

**“as open as possible,
as closed as necessary”**

([European Commission](#))

FAIR data does not have to be Open
however
Open data should always be FAIR

RDM to accomplish FAIR, open data

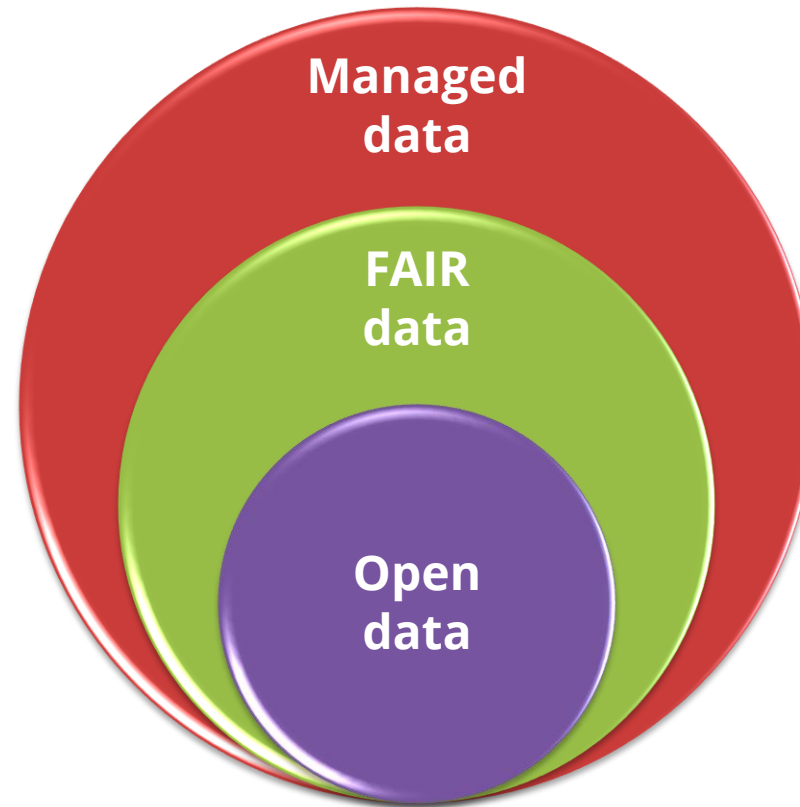
RDM is **necessary** for **useful** Open Data

FAIR principles **guide** RDM activities

FAIR data does **not** have to be **Open**

however

Open data should **always** be **FAIR**



Useful
for
me

Useful
for
others

Today

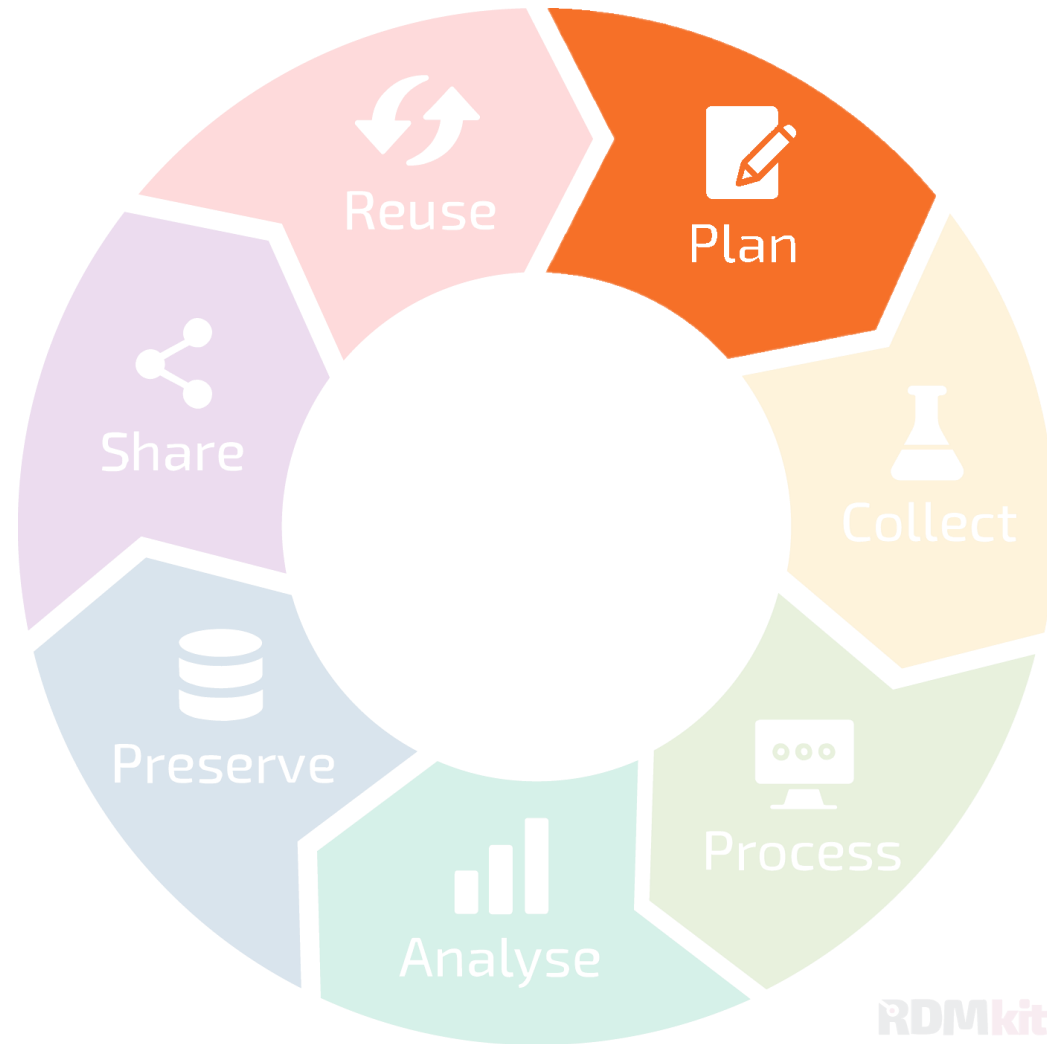
Key concepts

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- Research Data Management
- FAIR data
- Open data

Data management throughout the research data lifecycle

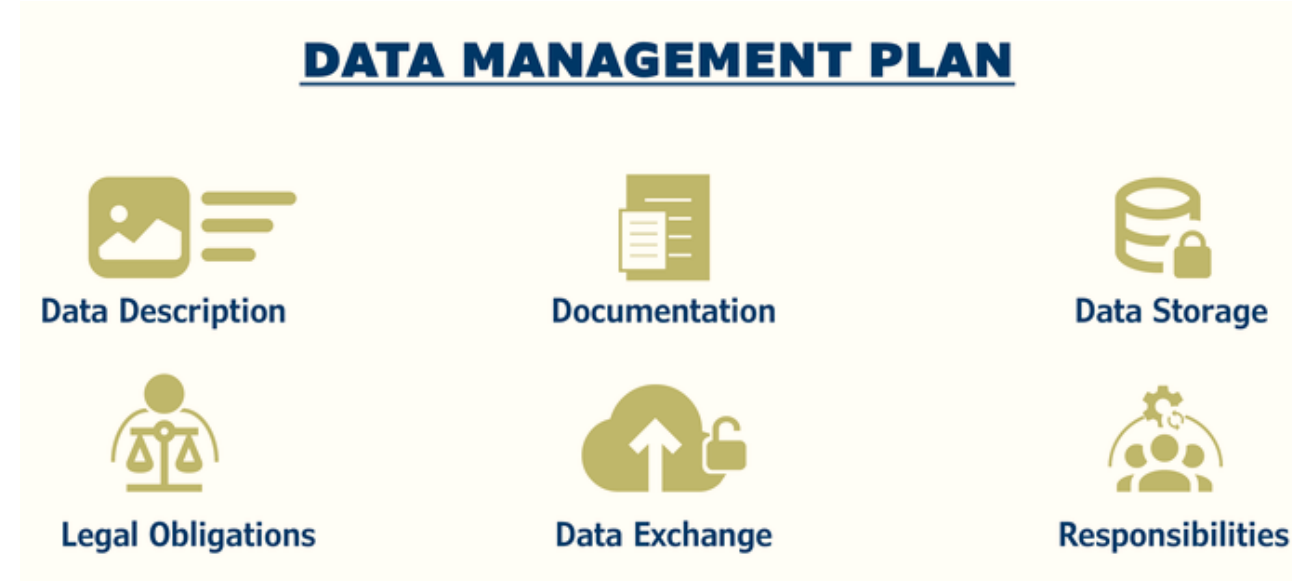
- Data Management Plans (DMP)
- Data collection
- Organizing your data
- Data documentation
- Storage
- Archiving
- Publishing data
- Data re-use

Data Management Plans



Data Management Plan (DMP)

- **Formal** document
- Describes your data
- Outlines all aspects of managing your data, both during and after your project
- It is a **living document**; can (*and should*) be continually **edited and updated**
- Helps make **RDM activities** more **concrete** and **actionable**
- Will save time, work, and potentially money



*image from [Data management plan](#)
Humboldt-Universität Berlin*

Did you know?

When you complete a draft of your DMP, you can submit it to RDM Support for review and feedback!

Example from DMP

1.2 If new data will be produced: describe the data you expect your research will generate and the format and volumes to be collected or produced.

Data description	Format	Volume	Conversion to open data format
Raw RNAseq data	diverse (fastq)	7 GB	
Processed RNAseq data	bam, bai, txt, csv	3 GB	txt, csv for processed data
Scans of plant architecture (raw and processed imaging data)	tiff, jpeg	180 GB	tiff
Textual and tabular documents	txt, csv, tsv	< 1GB	txt, csv, tsv
Scripts	r, py	< 1GB	can be opened as txt
Physical samples	freezer		

6.1 What resources (for example financial and time) will be dedicated to data management and ensuring that data will be FAIR (Findable, Accessible, Interoperable, Re-usable)?

0.05 fte (2h/week) will be spent by the employed technician to ensure that data comply with the FAIR principles.

Expected direct costs associated with data storage are <€1000 for 10 years.

Processing of the raw data will be done as a part of service included in the project costs.

DMP online

Project Details

Plan overview

Write Plan

Share

Download

expand all | collapse all

0/28

1. General features (0 / 2)

2. Data Collection (0 / 6)

3. Personal data (Data Protection Impact Assessment (DPIA) light) (0 / 6)

4. Data Storage and Backup (0 / 2)

5. Metadata and Documentation (0 / 2)

6. Data Analysis (0 / 1)

7. Data Preservation and Archiving (0 / 4)

8. Data Sharing Statement (0 / 5)

My Amazing Research Project

Project details

Contributors

Plan overview

Write Plan

Research outputs

Share

Request feedback

Download

expand all | collapse all

Data Collection (0 / 2)

What data will you collect or create?

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🔗

📅 ∨

Save

< Comments & Guidance

Guidance

Comments

DCC

Utrecht University

Questions to consider:

- What type, format and volume of data?
- Do your chosen formats and software enable sharing and long-term access to the data?
- Are there any existing data that you can reuse?

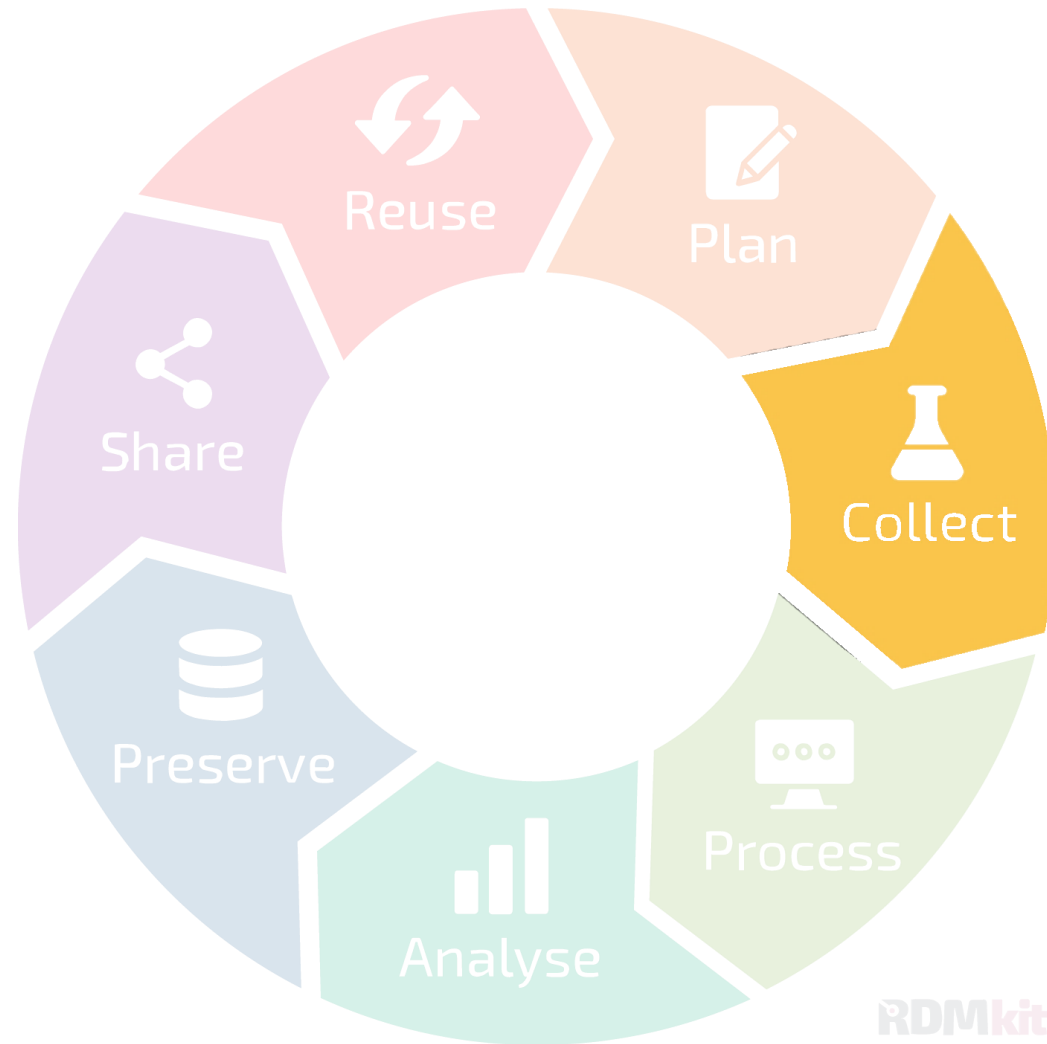
Guidance:

Give a brief description of the data, including any existing data or third-party sources that will be used, in each case noting its content, type and coverage. Outline and justify your choice of format and consider the implications of data format and data volumes in terms of storage, backup and access.

Use your institutional credentials to log into [DMPonline](#)

Data Collection

Preferred formats



Data collection table

Type	Description	Origin /collection	Formats	Software	Total file size	Number of files / samples
Biological data	Blood samples	Veterinarian	1mL/animal	NA	NA	250 animals
Lab and stable journals	Dates, protocols, lab worker, etc.	Lab worker / researcher	.csv and .txt	eLabjournal	100-500 MB	2 lab journals (consist of multiple files)
Lab results	Gene expression and antibody titers	Microarray data and ELISA data	.csv, .Rdata, .chp, .txt	Affymetrix, locally developed tool	200 GB	20 data output files
Behavioural data	Animal behavior visually scored	Researcher and research assistants	.csv	Noldus observer and ethovision	Kb's	2 output datafiles
Bodyweight	Biweekly bodyweight	Stable workers	.csv	NA	Kb's	1 output datafile
Statistical analyses	Scripts/codes and output tables and figures	Researcher	.R, .SAS, .csv, .tiff	R, Rstudio, SAS, Excel	1-50 MB	5 scripts, 5 table files, 5 figure files

Preferred formats

Preferred formats offer the best **long-term** guarantees in terms of usability, accessibility and sustainability.

Preferred Formats Characteristics:

- Non-proprietary / Open Source
- Unencrypted
- Uncompressed
- Interoperable
- And/or: commonly used

dans.knaw.nl/en/file-formats/

Text



Quantitative data



Images



Audio



Video



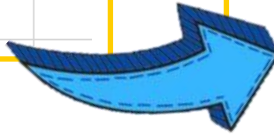
Interoperable file types

data_v01

Excel Workbook (*.xlsx)

 Save

	A	B	C
1	Letter	Animal	Number
2	A	Armadillo	443
3	B	Bat	444
4	C	Camel	445
5	D	Donkey	446
6	E	Elephant	447
7	F	Fox	448
8	G	Gecko	449
9	H	Horse	450
10	I	Iguana	451
11	J	Jaguar	452
12	K	Kangaroo	453
13	L	Lizard	454
14	M	Mole	455
15			



*data_v01 - Notepad

File Edit Format View Help

data_v01

CSV (Comma delimited) (*.csv)

 Save

Letter;Animal;Number
A;Armadillo;443
B;Bat;444
C;Camel;445
D;Donkey;446
E;Elephant;447
F;Fox;448
G;Gecko;449
H;Horse;450
I;Iguana;451
J;Jaguar;452
K;Kangaroo;453
L;Lizard;454
M;Mole;455
|

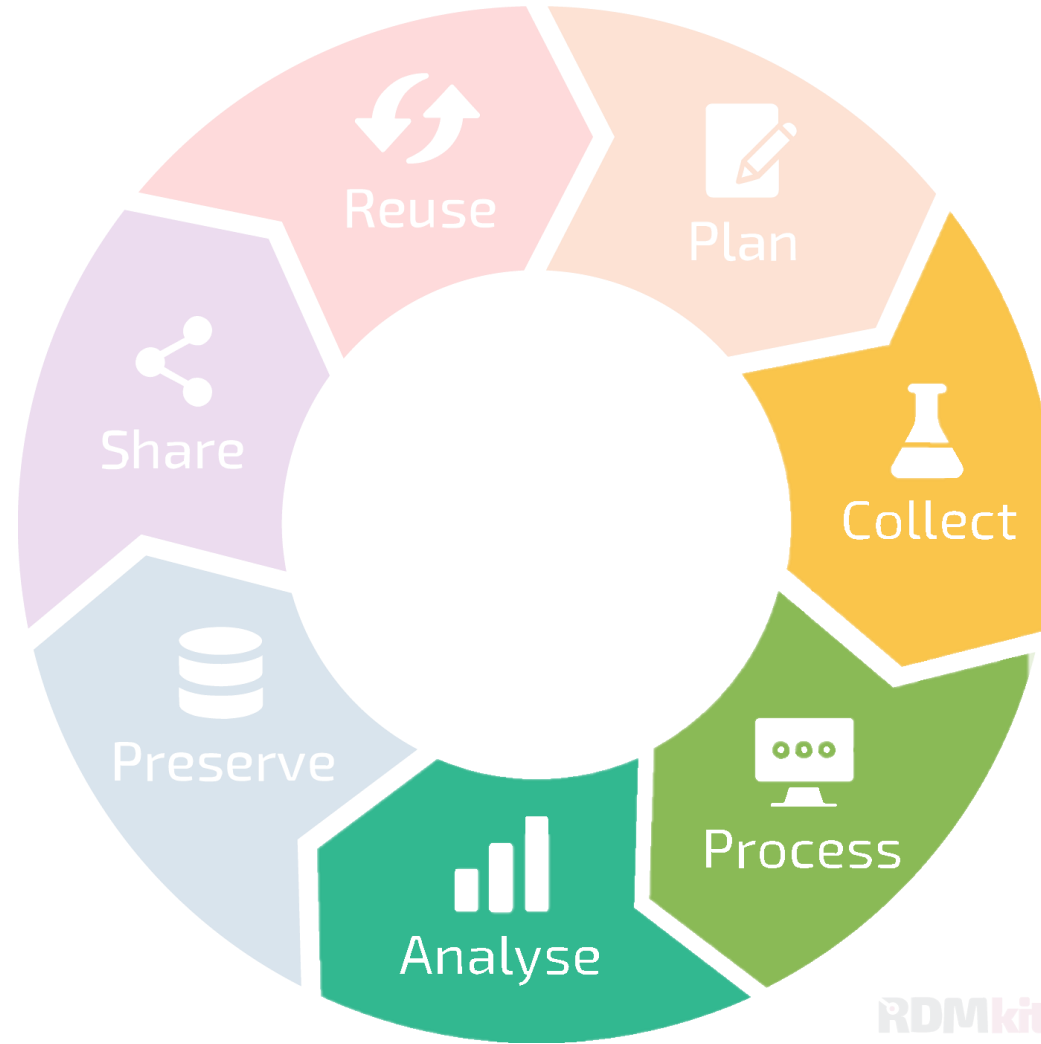
You might need to convert your files to a format suitable for **preservation**

Organising your data

Folder structure

File naming

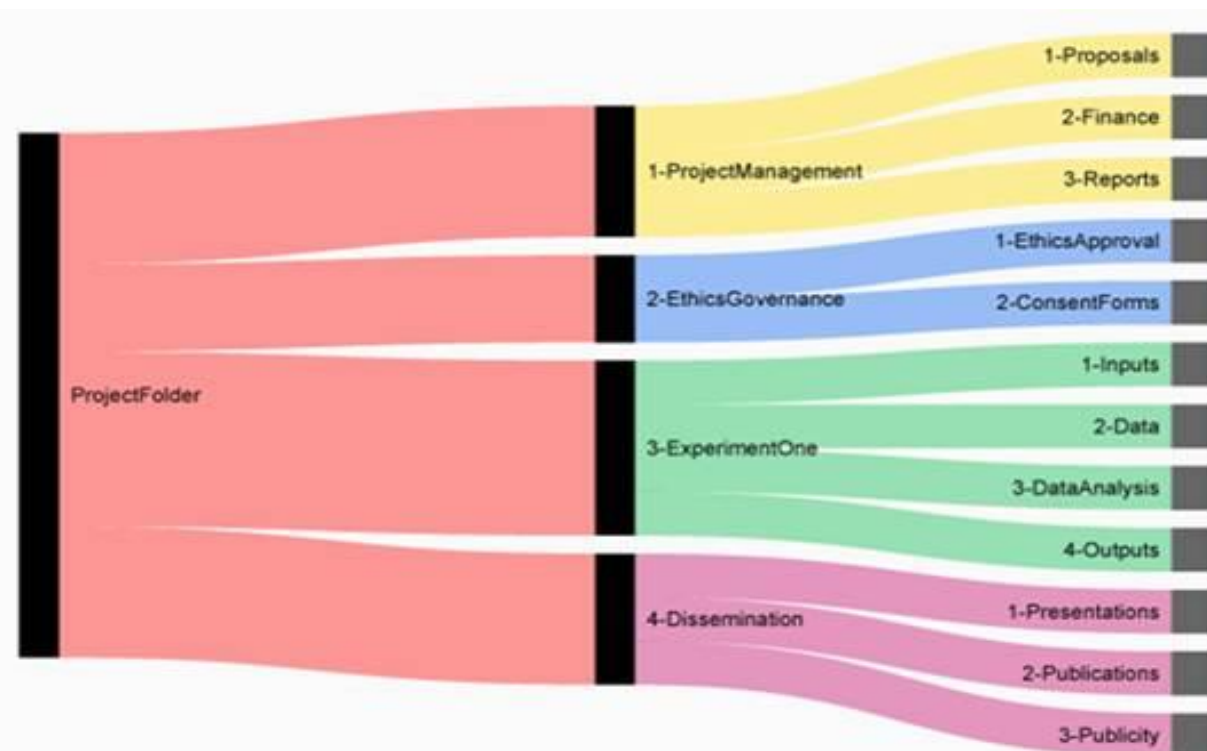
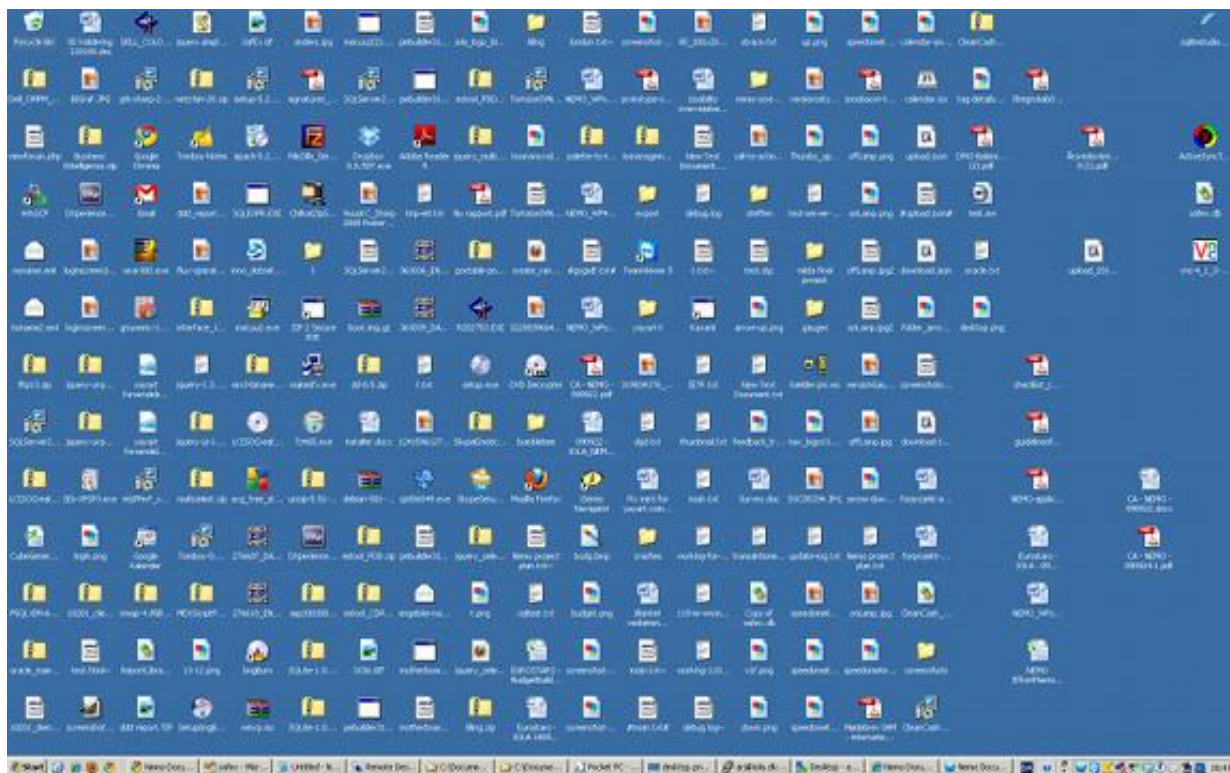
Version control



Does your desktop look like this?

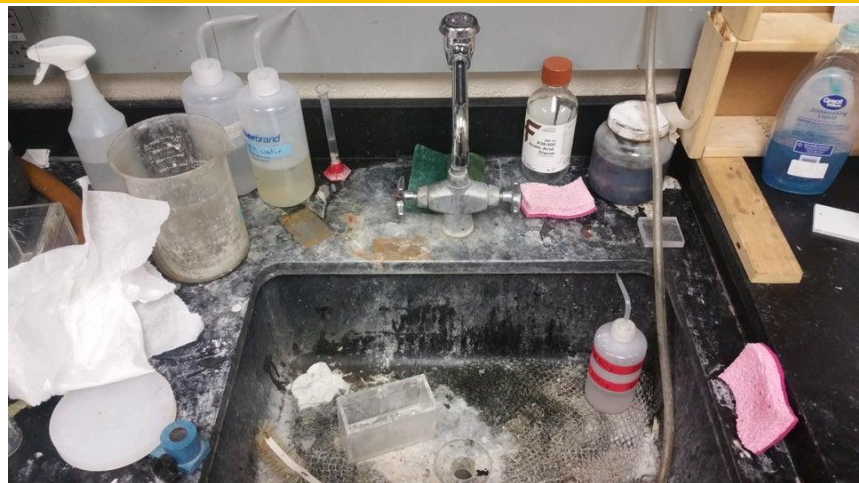


It may be time for some structure!



Example folder structure - From:
'Setting up an Organised Folder Structure for Research Projects'
Posted June 4, 2014 Blog by Nikola Vukovic

Keeping your data clean



.RData	4/8/2021 11:41 PM
.Rhistory	6/13/2021 10:48 PM
8.03.21 MS GUS	4/5/2021 5:37 PM
8.03.21	4/4/2021 12:22 AM
8.03	10/3/2021 8:45 PM
8.03	4/5/2021 11:46 AM
AIR GUS	4/5/2021 6:13 PM
AIR new	4/5/2021 9:51 PM
AIR scr	10/3/2021 8:42 PM
AIR	4/7/2021 12:02 PM
ANOVA Exercise	11/29/2020 9:50 PM
AnovaScr	12/16/2020 9:48 PM
Biomass vs Thousand grain weight	6/13/2021 10:23 PM
coursera	4/3/2021 11:50 PM
DATA	12/17/2020 8:19 AM
dataset.14.2020_2021	6/10/2021 9:55 AM
DATA	12/17/2020 8:19 AM



Name	
▼ Folder 1	
▼ Subfolder 1A	
▼ Subfolder 1B	
▶ Subfolder 1B-1	
▼ Folder 2	
▶ Subfolder 2A	
▶ Subfolder 2B	
▶ Subfolder 2C	
▼ Folder 3	

Research project package

Part of documentation: determining contents and organization of Research Project Package:

Data

- Raw data
- Processed data

Documentation and Metadata

- Variable codebook
- Metadata standard (at data and descriptive level; if any)
- Protocols and Standard operating procedures (SOP)

Scripts & Software

- Analysis
- Processing

Legal Documents

- Data protection impact assessment (DPIA)
- Licenses
- Informed Consent form (template only)
- Data transfer agreement (DTA)

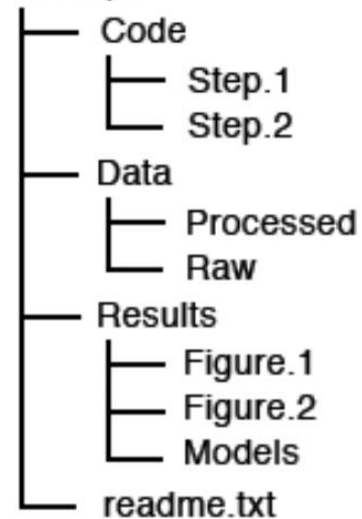
Administration

- Ethical review
- Grant & Consortium: applications and agreements
- Data management plan (DMP)

Directory building

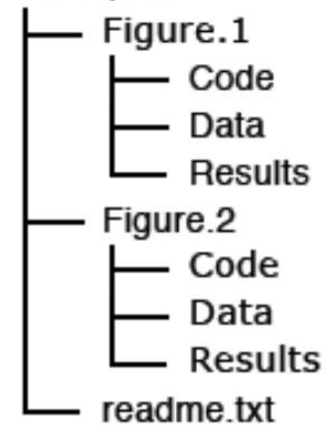
A) Organized by File type

Example.A



B) Organized by Analysis

Example.B



Folder structure can be based on

- Project
- Chronologically (month / year / sessions)
- Analysis method/equipment
- Data type
- Person generating the data/folder

Best Practices

- Determine in advance what your folder structure should be
- Make it logical - from time to time: 'check the logic'
- Describe your structure and files in a README file

Folder structure tools

Folder-structure automation

(after creating your folder structure template)

- [Organize: The file management automation tool](#) (open source)
- [File Juggler](#) (Windows, paid)
- [Hazel](#) (MacOS, paid)

Re-using templates:



[Yeoman](#) (open source)



[Cookiecutter](#) (open source)

File Renaming Tools

- [Simple Renamer](#) (open source, Mac)
- [Batch File Rename](#) (open source)
- [GPRename](#) (Linux, free)
- [Bulk Rename Utility](#) (Windows, free)
- [Microsoft PowerToys](#) (Windows, free)
- [Renamer](#) (MacOS, paid)
- [NameChanger](#) (MacOS, paid)

File naming

A file name can contain valuable **metadata** if information rich.
Consider using one or more of the following:

- Project or experiment name or acronym
- Location/spatial coordinates
- Researcher name/initials
- Date (range) of experiment
- Type of data
- Conditions
- Version number of file

This will differ depending on your project/experiment. Find a method that fits best for your data.
There is no “perfect method”!

File naming conventions

	Naming Convention	Example	Description
	original name	an awesome name	N/A
	snake_case	an_awesome_name	All words lowercase, separated by an underscore (_)
	kebab-case	an-awesome-name	All words lowercase, separated by a hyphen (-)
	PascalCase	AnAwesomeName	All words capitalized. Spaces not used.
	camelCase	anAwesomeName	First word lowercase, remaining words capitalized. Spaces not used.

File naming as part of Metadata

```
> stringr::str_split_fixed(flist, "[_\\.]", 5)
```

	[,1]	[,2]	[,3]	[,4]	[,5]
[1,]	"2013-06-26"	"BRAFWTNEGASSAY"	"Plasmid-Cellline-100-1MutantFraction"	"A01"	"csv"
[2,]	"2013-06-26"	"BRAFWTNEGASSAY"	"Plasmid-Cellline-100-1MutantFraction"	"A02"	"csv"
[3,]	"2013-06-26"	"BRAFWTNEGASSAY"	"Plasmid-Cellline-100-1MutantFraction"	"A03"	"csv"
[4,]	"2013-06-26"	"BRAFWTNEGASSAY"	"Plasmid-Cellline-100-1MutantFraction"	"B01"	"csv"
[5,]	"2013-06-26"	"BRAFWTNEGASSAY"	"Plasmid-Cellline-100-1MutantFraction"	"B02"	"csv"
[6,]	"2013-06-26"	"BRAFWTNEGASSAY"	"Plasmid-Cellline-100-1MutantFraction"	"B03"	"csv"



“_” underscore used to delimit units of meta-data I want later

“-” hyphen used to delimit words so my eyes don’t bleed

Version control

Version control is the process of managing changes to files.

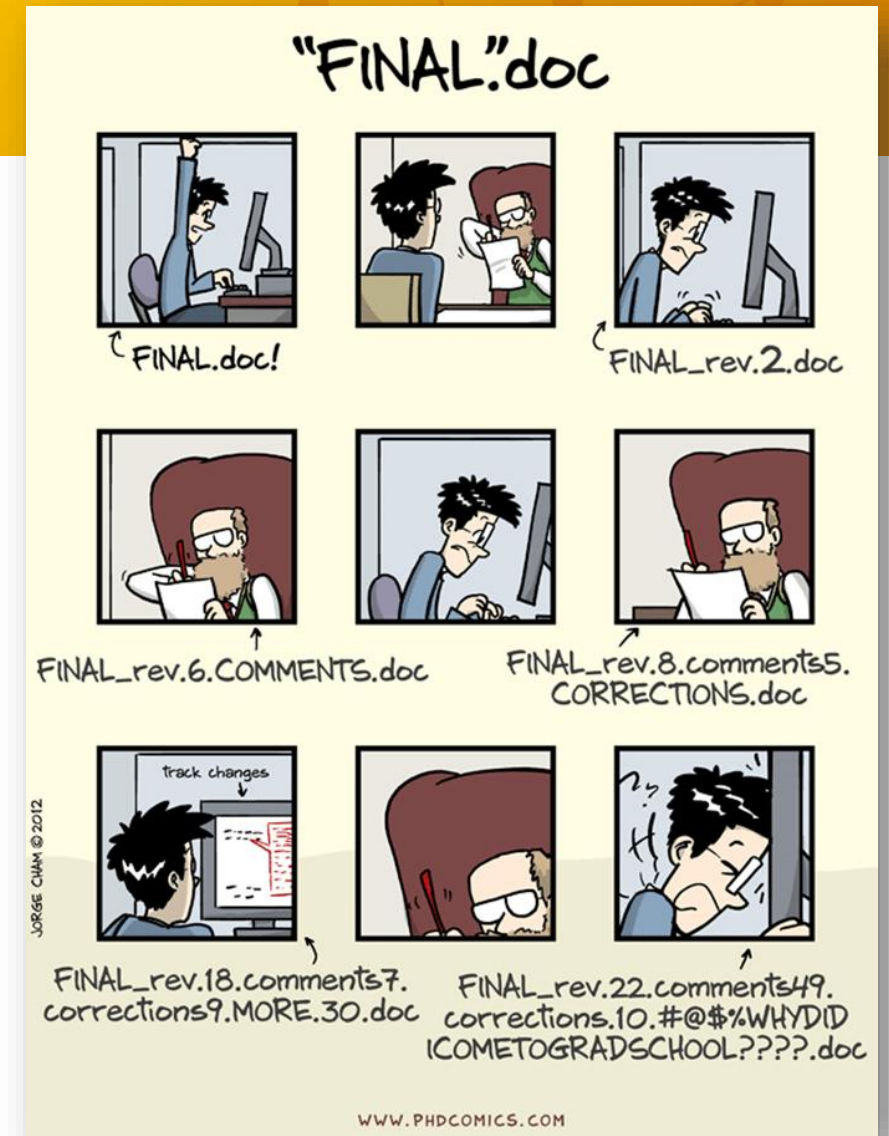
Manual version control:

as simple as adding a version number or a date (using YYYYMMDD format for sorting purposes) to the end of the file name.

Examples:

ASIST_abstract_v3.docx

ASIST_abstract_20250522.docx



Human- and machine-readable naming

Human-Readable

- Informative of content
- Preferably <25 characters long
- **Combine** naming conventions:

2026-01-21_CytAssay_Plasmid-CL100-1MF_A01.csv

Machine-Readable

AVOID:

- **Spaces**
- Special characters like \$ @ % # & * ! / \
- Punction characters like , : ; ' "
- Accented characters like á ü ñ

Numbering

Sorting by logical structure => **number at start** of file name

- Use **multiple digits** as 01 or 001 depending on number of expected files

Dates

- Sorting chronologically => **date at start** of file name
- Dates by ISO 8601 standard:

YYYY-MM-DD or **YYYYMMDD**

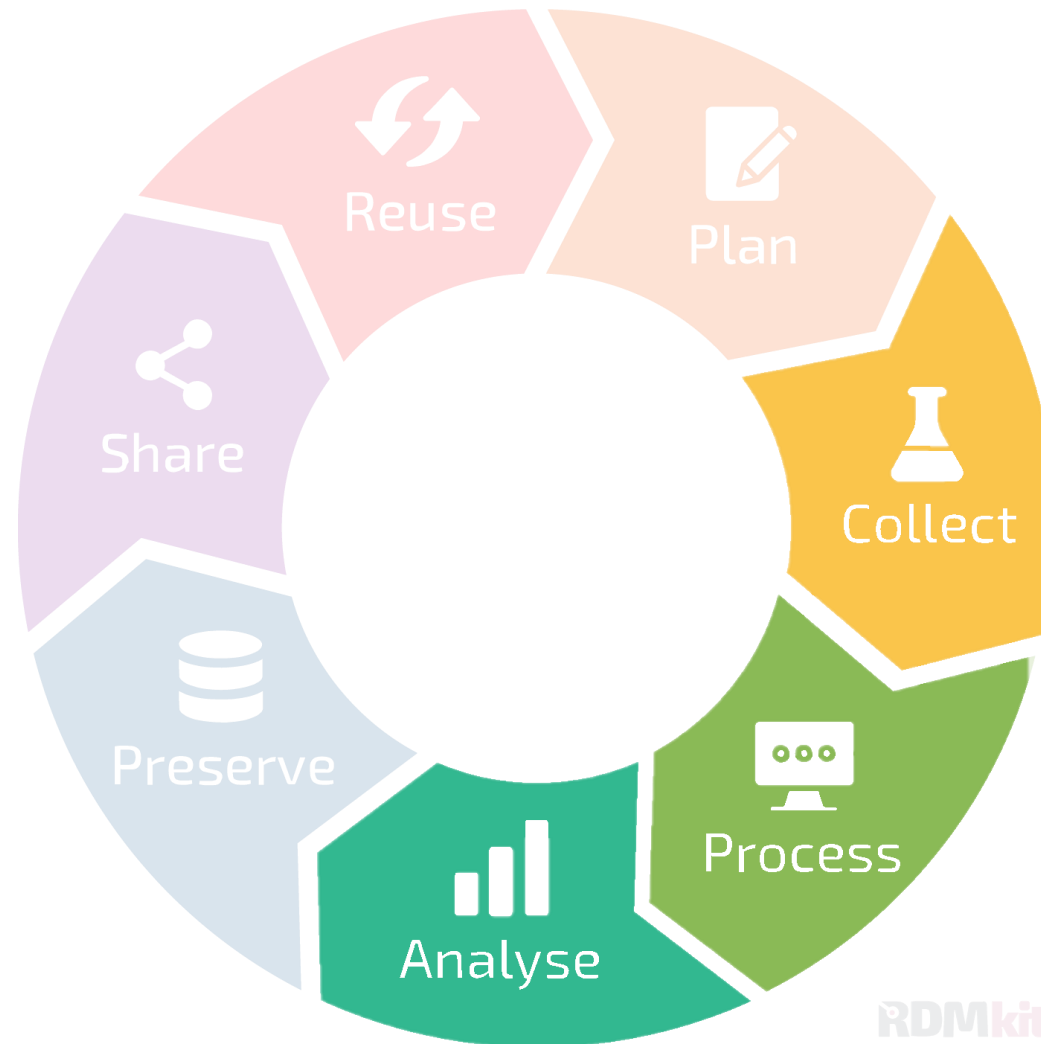
Versioning

- Specify versions using
ordinal numbers for **major revisions** (1, 2, 3)
decimals for **minor changes** (1.1, 1.2, 2.1, 2.2)
- Alternatively use multiple digits (v01 and v02)

Quick Break

Data Documentation

*Documentation
Metadata*



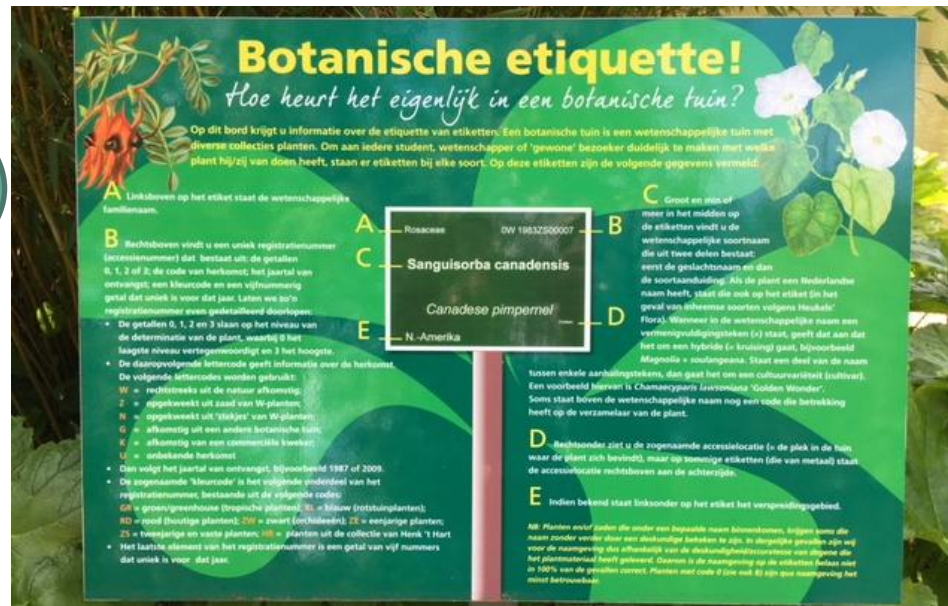
Documentation: Leaves, Labels & Legends



0 G 2009 GR 01628

0 = level of identification (0-2)
G = from other bot. garden
2009 = year received
GR = Green House
01628 = unique number

Explanation
(in Dutch) →



Documentation

- Refers to **contextual information** pertaining to your research data
- Meant to be **human-readable**
- Crucial aspect of **interoperability** and **reusability**
- Accompanies (structured) metadata
- Guides users to understand and interpret your data and reuse it effectively

Examples

- **README files**
- **Codebooks**
- Lab notebooks
- Data Management Plan (DMP)
- Study protocol / methodology
- Grant / study proposals
- Legal / policy / administrative documents

Documentation Checklist!

Here is a starter checklist (v0) that you can use to make an inventory of your documentation:

<https://tinyurl.com/documentation-checklist>

Documentation: README file

- Information about data or software
- Ensures data can be correctly interpreted and used
- Shared with open dataset
- Describes methods used for data collection and analysis and include data/software-specific information



[GitHub - UtrechtUniversity/README-Templates](https://github.com/UtrechtUniversity/README-Templates)

[Make a README](#)

Best practices

- Create one readme file for each dataset
- Name the file README; not readme, read_me, ABOUT, etc.
- Write your readme document as a plain text file; save as README.txt or README.md when writing in Markdown.

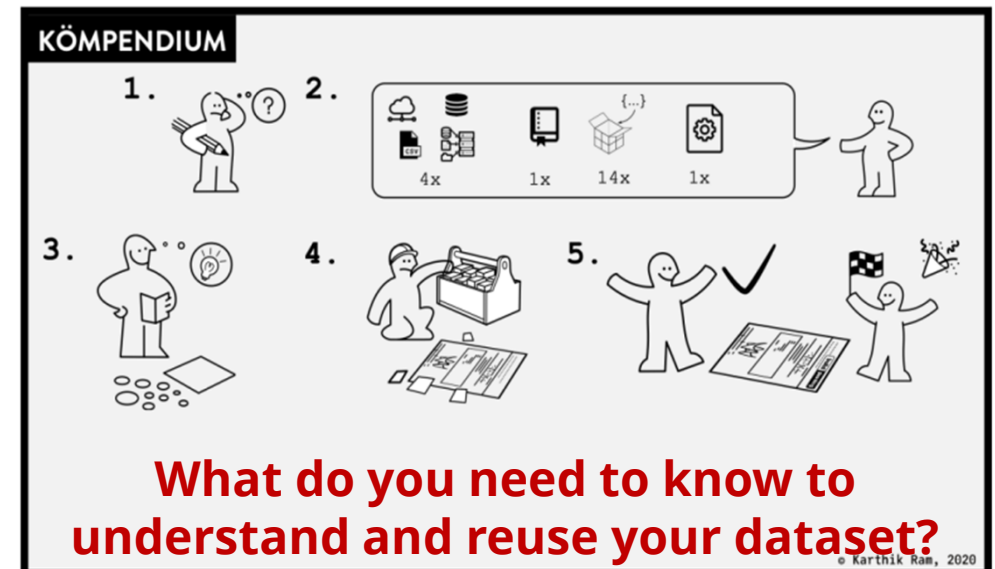
Minimum content

- Title of dataset
- For each file / group of similar files: short description of what data it contains
- Method description for collecting/ generating/ processing the data
- Full names and definitions of column headings
- Units of measurement
- Definitions for codes or symbols used to record missing data
- Specialized formats or abbreviations used

Metadata on your project & data

What is metadata?

- Structured information that describes one or more aspects of your research data.
- E.g. content, quality, format, location, access rights
- In other words, metadata = 'data about data'
- Typically machine-readable.



Metadata – Data about data

DATA-LEVEL METADATA

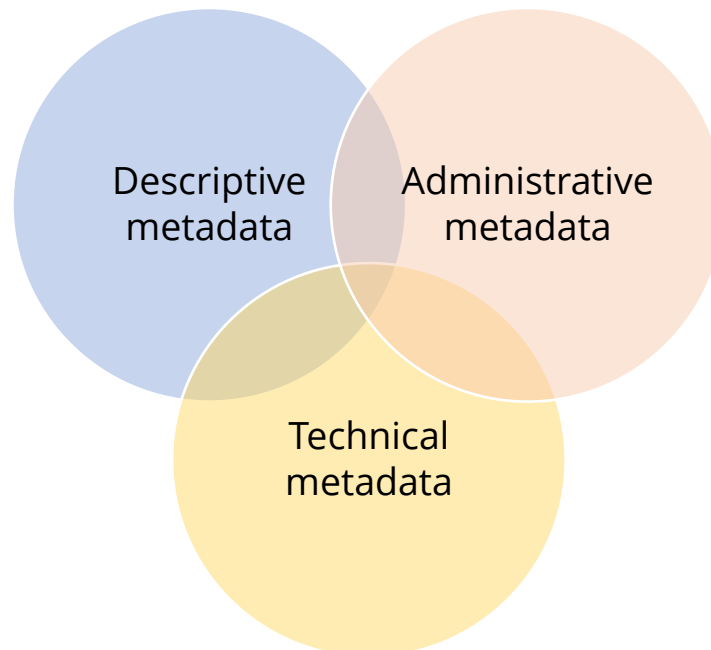
- Describes individual datasets or files
- May differ between datasets
- Enables interpretation, validation, and reuse

- Dataset title and description
- Data creator(s)
- Date of data collection
- File format and size
- Variables and definitions
- Units of measurement
- Processing steps and versions
- Codebook

PROJECT-LEVEL METADATA

- Describes the research project as a whole
- Applies to all datasets
- Provides context, ownership, and governance

- Project title and description
- Principal investigator(s)
- Institution(s) and collaborators
- Funding agency and grant number
- Research objectives
- Ethical approval
- Data sensitivity and access conditions



Metadata Standards

Descriptive level

- Descriptive Metadata Standards are quite common.
- Often defined by the repository where you will be preserving or sharing your data.
- Common ones: **Datacite 4.1 & Dublin Core**

Dataverse and Yoda both use Datacite 4.1 as their descriptive metadata standard.

The 15 Dublin Core Metadata Elements

ISO 15836 Standard for Resource Description

 Title The name given to the resource <i>A Pilot's Guide to Aircraft Insurance</i>	 Creator The entity primarily responsible for creating the content <i>Shakespeare, William</i>	 Subject The topic of the resource, expressed as keywords or codes <i>Aircraft leasing and renting</i>
 Description A summary or abstract of the resource <i>Illustrated guide to airport markings and lighting signals...</i>	 Publisher The entity responsible for making the resource available <i>MIT Press</i>	 Contributor An entity contributing to the resource <i>Jane Smith</i>
 Date A point or period of time in the resource lifecycle <i>1998-02-16</i>	 Type The nature or genre of the content <i>Image or Text</i>	 Format The physical or digital form of the resource <i>image/jpeg or 40 x 512 pixels</i>
 Identifier A unique reference to the resource <i>ISBN:0385424728</i>	 Source A related resource from which the described resource is derived <i>Image from page 54 of the 1922 edition of Romeo and Juliet</i>	 Language The language of the resource <i>en or eng</i>
 Relation A reference to a related resource <i>IsPartOf: Two Lives</i>	 Coverage The spatial or temporal scope of the resource <i>17th century or Boston, MA</i>	 Rights Information about access or usage restrictions <i>Access limited to members</i>

Minimum Reporting Requirements

- Domain specific standards of metadata to be provided
- Often determined by repository
- List of required and suggested minimum metadata fields

Life Sciences examples:

Minimum **I**nformation **A**bout a **M**icroarray **E**xperiment (MIAME)

Minimum **I**nformation about Any **S**equence(MIxS)

Functional **A**nnotation of **A**nimal **G**enomes (FAANG)

Metadata field	Required?	Definition	Comment
unique ID	required	Identifier for the assay that is at least unique within the project	
experiment type	required	The type of experiment performed, eg ATAC-seq or seqFISH	ontology field - e.g. EFO or OBI
extracted nucleic acid/material type	required	The type of material that was extracted from the sample, eg polyA RNA	ontology field - e.g. ChEBI or EFO
platform	required	The type of instrument used to perform the assay, eg Illumina HiSeq 4000 or Fluidigm C1 microfluidics platform	ontology field - e.g. EFO or OBI
nucleic acid extraction method	required	Technique used to extract the nucleic acid from the cell	ontology field - e.g. EFO or OBI

Codebooks

- Example of **data-level** metadata
- Explains **meaning** of variable names and values
- **Human-readable** codebook (usually word/pdf file, tables)
<https://osf.io/2hkrw>
- **Machine-readable codebook** (usually csv or xlsx) can be used by programming tools
Cookbook for a Codebook:
<https://osf.io/q3s4q/>

Information to include:

- Variable Names
- Readable Variable Name
- Measurement Units
- Allowed Values
- Definition Of The Variable
- Synonyms For The Variable Name (Optional)
- Description Of The Variable (Optional)
- Other Resources

Example Codebook

data/		Sheet_1		
OSF Storage (United States)		Show rows with cells including:		
0.1 datadictionary.csv		Variable	Variable name	Mesaurement unit
0.2 raw.pdf		Participant ID number	ID	Numeric
0.3 clean.csv		Group number	GROUP	Numeric
		Age in years	AGE	Numeric
		Date of birth	DOB	mm/dd/yyyy
		Gender	SEX	Numeric
		Date of survey	SURVEY	mm/dd/yyyy
		Self-reported consumer spending	SPEND	Numeric
		Market sentiment	SENTIMENT	Numeric
		Actual GDP growth	GDP	Numeric
				Allowed values
				Description
				001-999
				ID number assigned to participant in sequential order
				1-30
				Group assigned to participant based on ID number
				18.0-65.0
				Age of participant in years
				1-12/1-31/1951-1998
				Participant's date of birth
				1 = male 2 = female
				Participant's gender
				01/01/2015 – 01/01/2016
				When the participant completed the survey
				0-100,000,000
				Self-reported average yearly expenditure
				1 = negative 2 = neutral 3 = positive
				Sentiment towards US domestic economy
				-5.0-5.0
				Average US yearly GDP growth

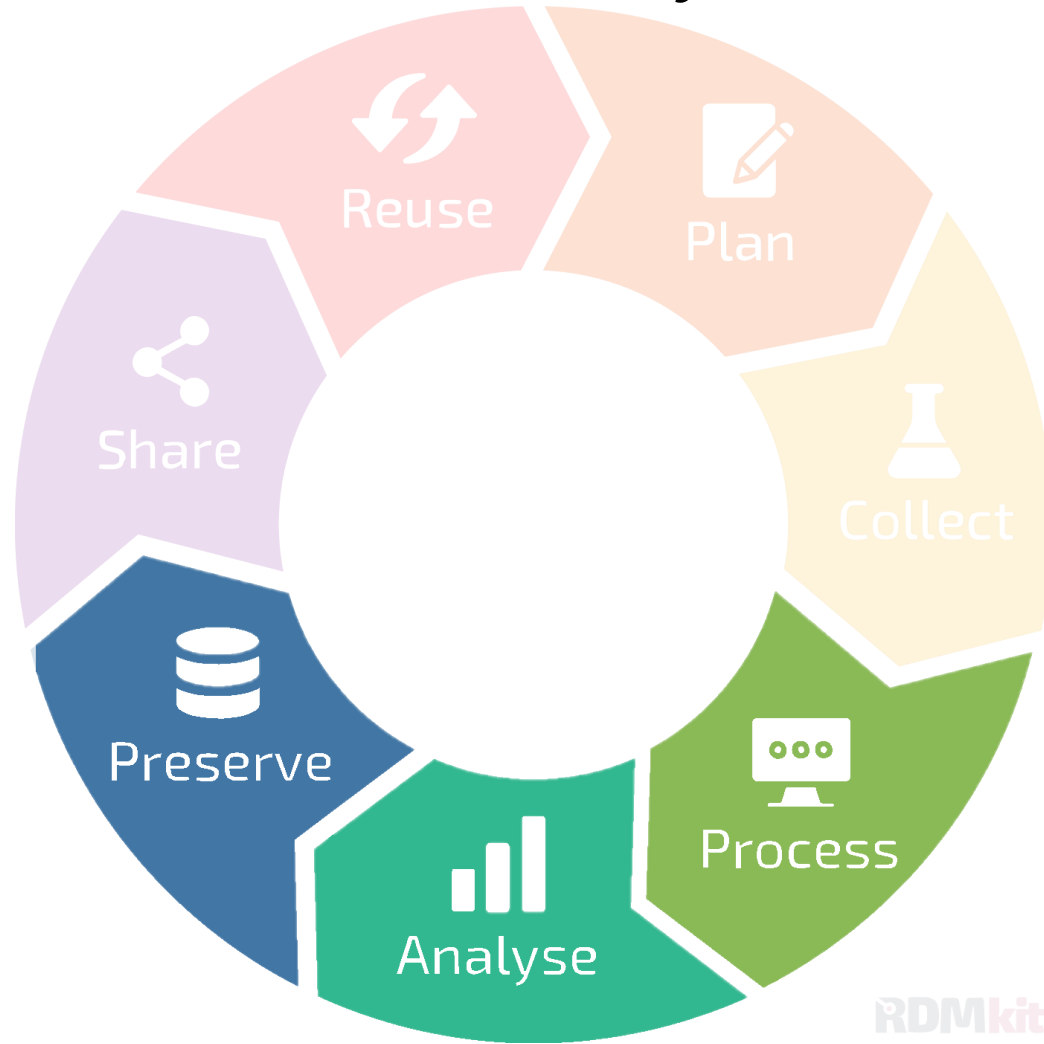
[How to Make a Data Dictionary - OSF Support](#)

Data Storage

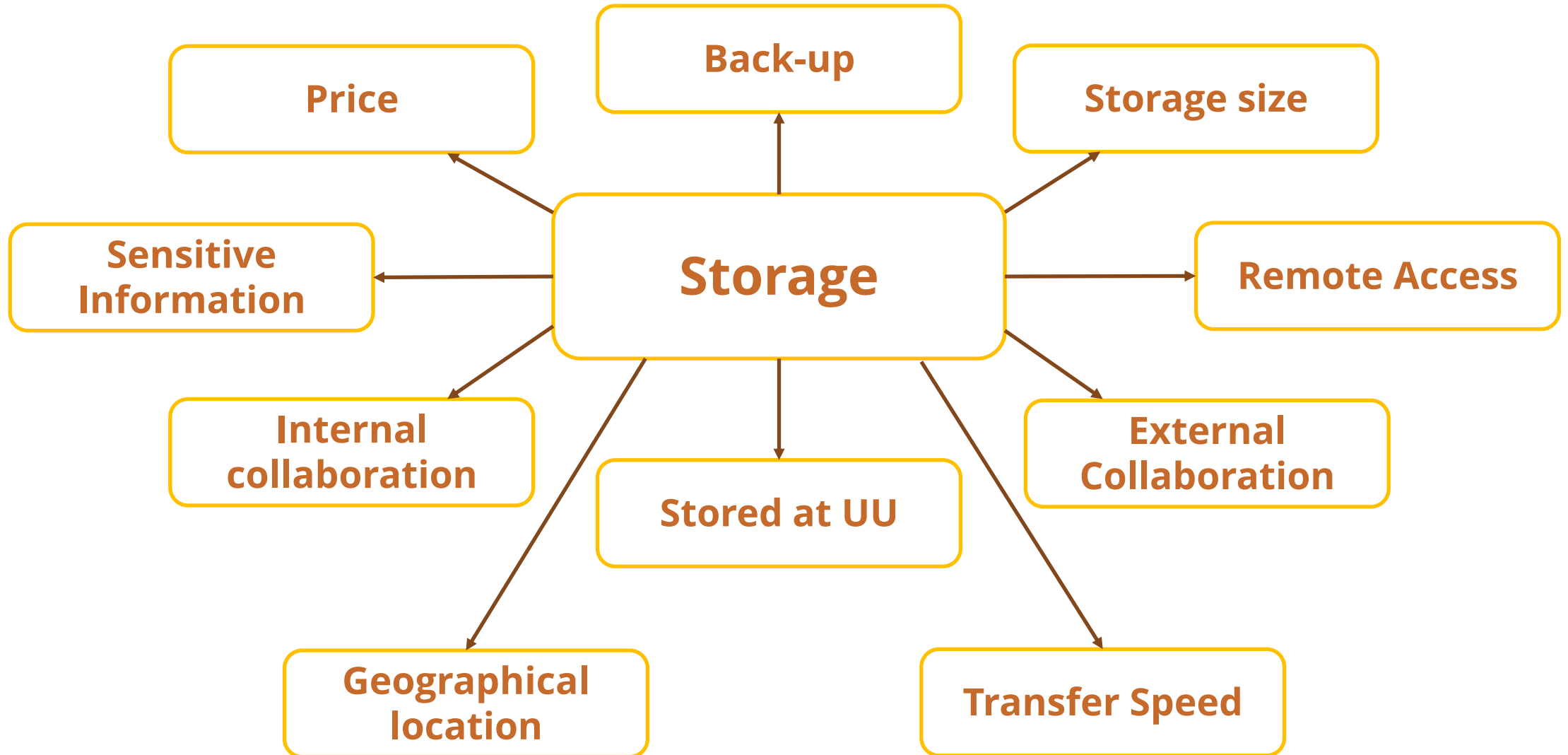
Storage solutions

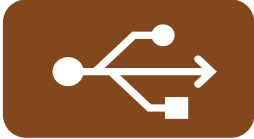
Personal data

Data security



What do you consider when choosing a storage solution?



Storage Option					YODA			
Storage size	Varied	Varied	Varied	Varied	Varied	Varied	1TB	1TB
Price	NA	NA	Faculty	Faculty	TB €4/m	UU	UU	UU
Back-up	✗	✗	✓	✓	✓	✓	✓	✓
Controlled by UU	✗	✗	✓	✓	✓	✓	✓	✓
Internal collaboration	✗	✗	✗	✓	✓	✓	✓	✓
External Collaboration	✗	✗	✗	✗	✓	✓	✓	✓
Sensitive Information	✗	✗	✓	✓	✓	✓	✓	✓
Remote Access	✗	✗	✓	✓	✓	✓	✓	✓

Storage solutions



Use the [Data Storage Finder](#) to select an appropriate solution!

Read more about **information security** at the UU on our intranet: [Information Security - Safe tooling and software](#)

More information: RDM Support guides on [Storing Data](#) and [Preserving data](#)

BEST PRACTICES IN STORING DATA

I. Choose storage media wisely



II. Manage versions and copies of your data carefully



III. Structure names and folders



IV. Find and understand your data by assigning metadata



V. Use standard file formats



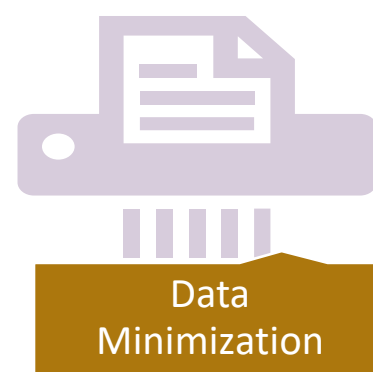
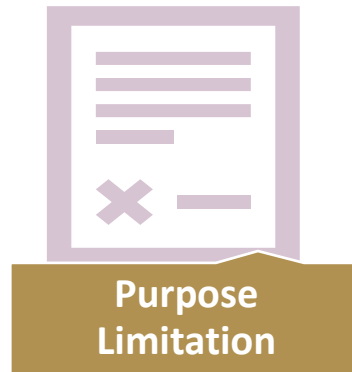
VI. Secure your data files



Think about a storage and back-up strategy for both RAW and PROCESSED DATA.

Backup a second location until the project is completed or the files are uploaded to a repository!

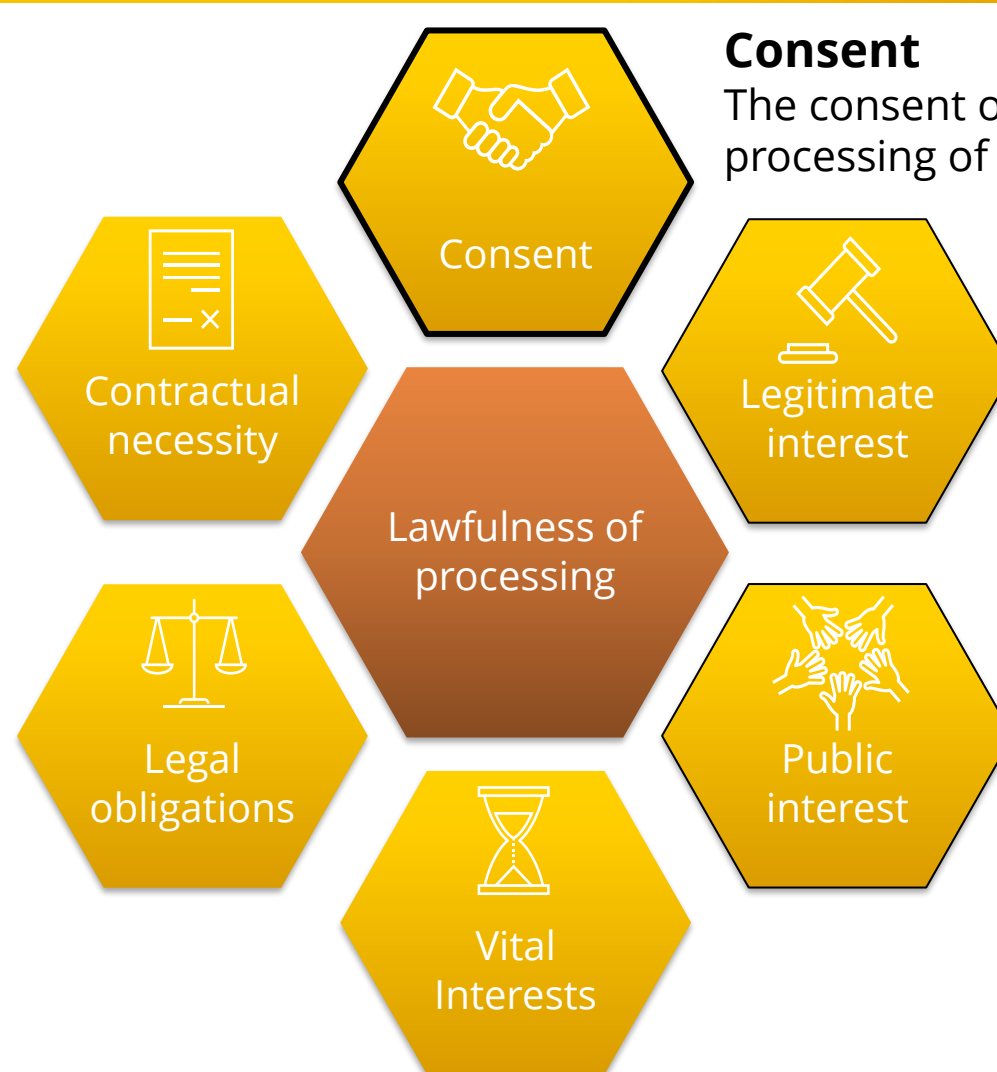
Personal data: the 7 principles of the GDPR



Lawfulness of Processing (Art. 6)

Personal data may only be **processed** if at least one of the following applies:

Note:
Looking at data,
storing data,
analyzing data
and even
deleting data
*are all forms of **processing**!*



Most commonly used

Less commonly used

Data security: three levels

Security of data files

- **Encryption**
- **Pseudonymization**
- Minimization
- Abstraction
- **Anonymization**
- Access control
- Secure transport
- Complete deletion

Security of computer system

- Firewall
- Antivirus software
- Installing updates
- Using secured WIFI
- Password-protected
- Device encryption

Physical data security

- Key & Lock
- Don't leave unattended
- Safe transport

Privacy and security measures



Encryption

- Makes data unreadable/inaccessible without a **password**
- Different levels of encryption
 - **Data encryption** – data file itself encrypted
 - **Drive encryption** – hard drive, where data file is located, is encrypted



BitLocker

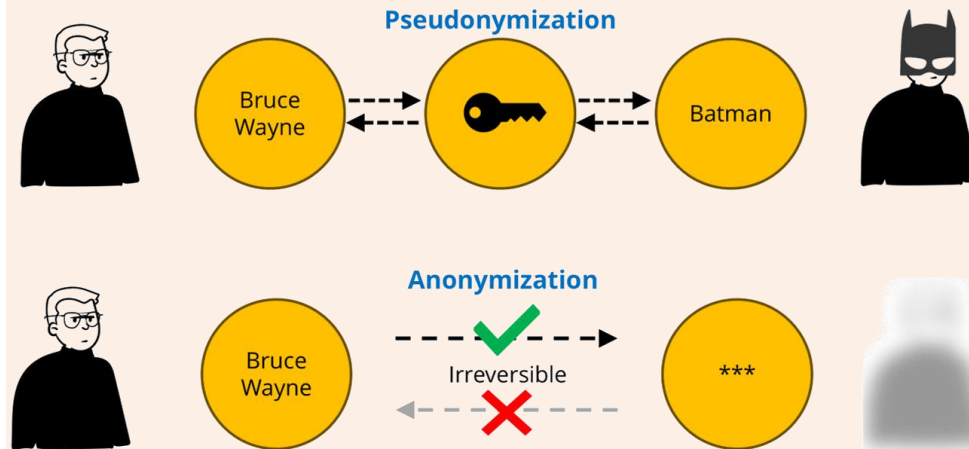


FileVault



Depersonalizing data

- **Pseudonymization**
- **Anonymization**
- Minimization
- Aggregation & Abstraction



Restricted Access

- During research
- After research
 - **Who** should have access to your data?
 - **How will you restrict access?**
 - **How will you enable access** to those authorized?
 - Where will you **describe** who gets to access the data?

More information

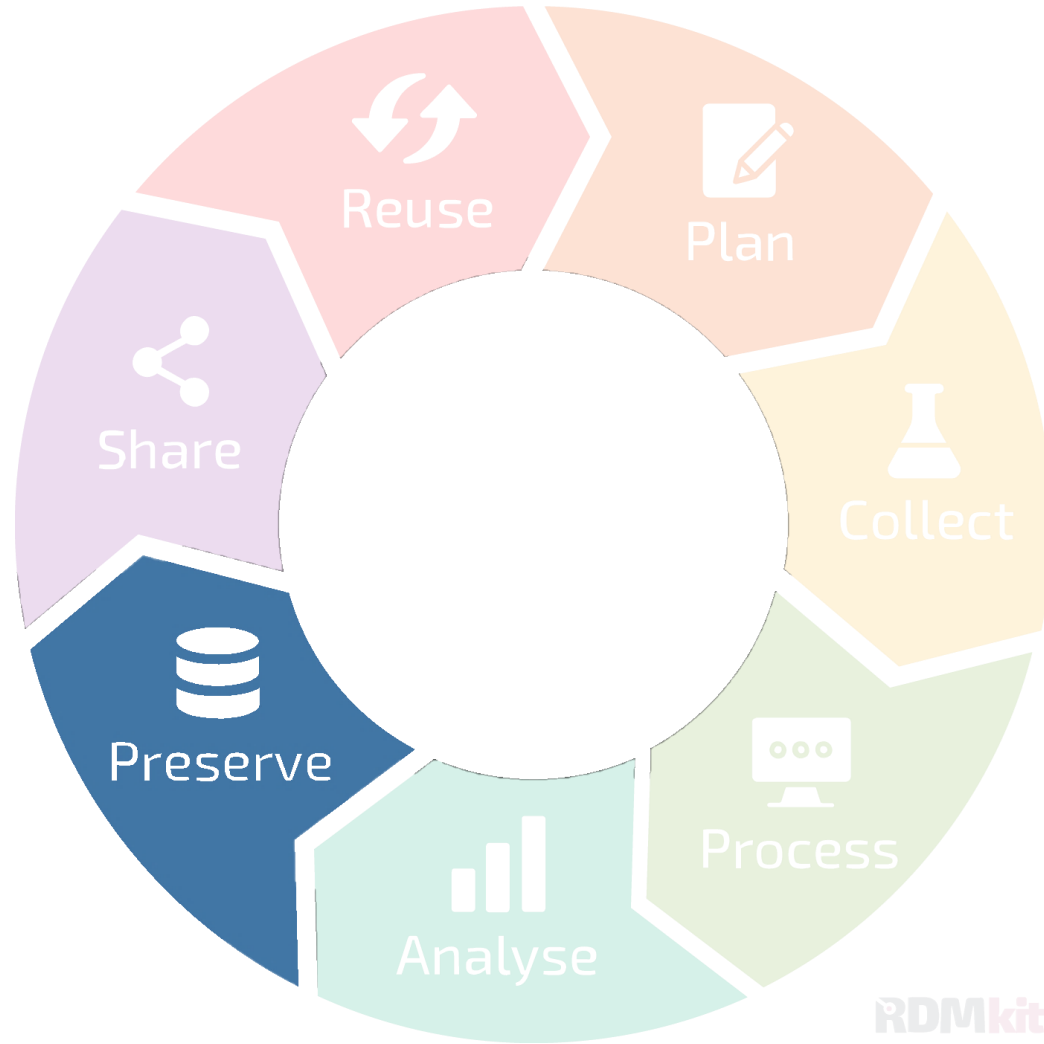
Data Privacy Handbook

Privacy basics for researchers (e-learning)

Handling Personal Data in Research Workshop



Data Archiving



Archiving data

Why archive?

- Promotes **scientific integrity** (verification)
- Enables (internal) **reuse**.
- Adds to the **findability** and **usability** of your research

Archiving research data includes:

- Ensuring **integrity**
- Long-term **availability**
- **Metadata** and **documentation**

UNIVERSITY POLICY FRAMEWORK FOR RESEARCH DATA (UU)

Raw data from research must be kept available for **a minimum of ten years** for verification purposes

...commencing from the date that the research results are published!

Read more about this in the policy framework:

www.uu.nl/sites/default/files/university_policy_framework_for_research_data_utrecht_university_-_january_2016.pdf

Data archiving package



Data

- Raw data
- Processed data

Documentation and Metadata

- Variable codebook
- Metadata standard (at data and descriptive level; if any)
- Protocols and Standard operating procedures (SOP)

Scripts & Software

- Analysis
- Processing

Legal Documents

- Data protection impact assessment (DPIA)
- Licenses
- Informed Consent form (template only)
- Data transfer agreement (DTA)

Administration

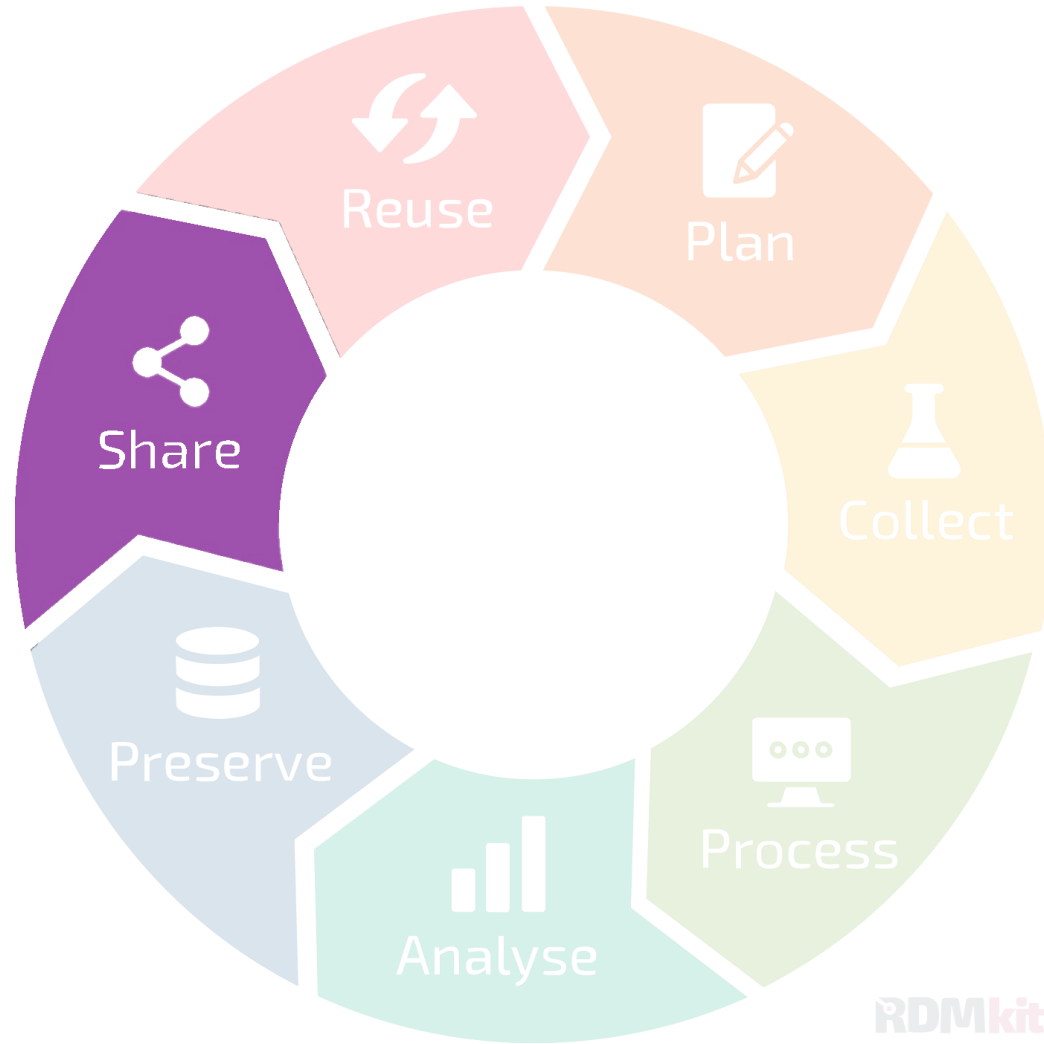
- Ethical review
- Grant & Consortium: applications and agreements
- Data management plan (DMP)

Data Publication

Publishing data

Repositories

Licensing



Publishing data

Why publish?

[Utrecht University Data Publication Guidelines](#)

[Publishing and sharing data](#)

- Promotes **scientific integrity** (verification)
- Follow **grant & journal mandates**
- Enables **reuse**.
- Adds to the **findability** and **usability** of your research
- Promotes **future collaborations & discovery**

“NWO introduced its research data management policy which aims to make research data generated as part of NWO funded projects as open and FAIR (...) as possible.”

Data publication package



Data

- Raw data 
- Processed data

Documentation and Metadata

- Variable codebook
- Metadata standard (at data and descriptive level; if any)
- Protocols and Standard operating procedures (SOP)

Scripts & Software

- Analysis
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Administration

- Ethical review
- Grant & Consortium: applications and agreements
- Data management plan (DMP)

Choosing the repository



1 DOMAIN-SPECIFIC REPOSITORY:

Use a trusted repository already established for your research domain.



2 INSTITUTIONAL OR RECOMMENDED DATA REPOSITORY:

If a trusted domain-specific repository is not available, use an institutional research data repository.

The UU supports Yoda & DataverseNL.



3 GENERAL PURPOSE REPOSITORY:

If neither of the above is available, use a general purpose repository like DANS EASY, Zenodo, 4TU.Center, Figshare.



UU Repository Finder

To take into account:

- Persistent identifier (**PID**)
- Is **long-term preservation** guaranteed or not?
- What are the **costs** per dataset or gigabyte?
- What is the physical storage **location** of data? (data protection laws)
- Does the repository allow for **restricted access**?
- Does the repository allow me to **choose the re-use licence**?
- Is the repository **certified**?

Tool to select an appropriate solution:
[Data Repository Finder \(uu.nl\)](https://uu.nl/data-repository-finder)

Data Repository Finder

This Data Repository Finder can help you choose the data repository that fits your needs when sharing and publishing your research data. For more information you can consult the Guide Publishing and sharing data or contact RDM Support.

1. Are you looking for a repository managed by Utrecht University? ⓘ

☐ Yes ☐ Not necessarily

2. Will you publish data under access restriction? ⓘ

☐ Yes ☐ Not necessarily

3. Are you looking for a repository with the option to deposit for free? ⓘ

☐ Yes ☐ Not necessarily

4. Do you wish to have the freedom to choose your own data usage licence? ⓘ

☐ Yes ☐ Not necessarily

5. Do you want to be able to create a private URL? ⓘ

☐ Yes ☐ Not necessarily


4TU.RESEARCH DATA


DATAVERSE NL


MENDELEY DATA


YODA


DANS DATA STATIONS


DRYAD


OPEN SCIENCE FRAMEWORK (OSF)


ZENODO

Open Data

“ Open Data is data that can be **freely used, re-used and redistributed by anyone** subject only, at most, to the requirements to attribute and share-alike ”

([The Open Data Handbook](#))

F A I R

Make your data...

**“as open as possible,
as closed as necessary”**

([European Commission](#))

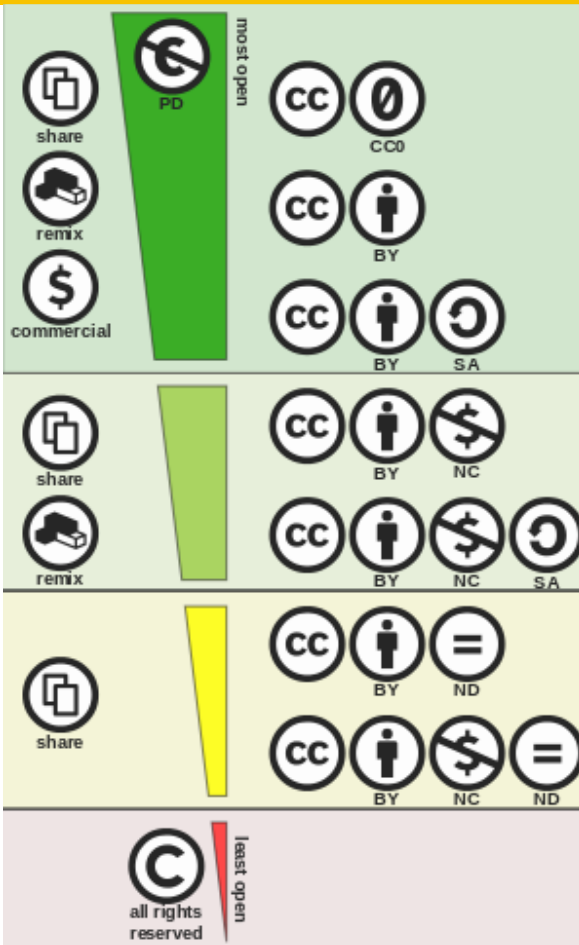
Reasons data can be FAIR yet not open:

- ☐ Ethical or cultural obligations
- ☐ Safety and security risks
- ☐ Contractual/licensing constraints
- ☐ National security/export controls
- ☐ Protection of sensitive ecological or heritage sites
- ☐ Embargoes for ongoing research
- ☐ Economic or political sensitivity
- ☐ Risks to vulnerable communities

FAIR data does not have to be Open
however
Open data should always be FAIR



Licensing of data



A license states **what a user is allowed to do** with your data and creates clarity and certainty for potential users.

Creative common licences are easy to use but **not the most suitable for data**

Stappenplan gebruik Creative Commons-licenties
<http://creativecommons.org>

Custom License

- To adequately protect your data from unwanted use: recommended to use a **custom license** instead of a creative commons license.
- Make data available under **restricted access**
- Access only granted after a **contract** with the adequate terms and conditions has been signed by the requestor.
- Very commonly used when dealing with personal data (Data Transfer Agreement).

[Licenses — Open Data Commons:
legal tools for open data](#)

 **Open Data Commons**
Legal tools for Open Data

Open Database Licence

You are free:

 *To Share:* To copy, distribute and use the database.

 *To Create:* To produce works from the database.

 *To Adapt:* To modify, transform and build upon the database.

As long as you:

 *Attribute:* You must attribute any public use of the database, or works produced from the database, in the manner specified in the ODbL. For any use or redistribution of the database, or works produced from it, you must make clear to others the license of the database and keep intact any notices on the original database.

 *Share-Alike:* If you publicly use any adapted version of this database, or works produced from an adapted database, you must also offer that adapted database under the ODbL.

 *Keep open:* If you redistribute the database, or an adapted version of it, then you may use technological measures that restrict the work (such as DRM) as long as you also redistribute a version without such measures.

AI policies

Guiding Principles

1. **Human accountability and human oversight:** Always remain the 'human in the loop' when working with GenAI. Ensure content generated by AI is factually accurate.
2. **Data protection and security:** Familiarise yourself with the UU data security policies and know how to find expert help on data security in the organisation.
3. **Transparency; show and tell:** Proactively learn about the benefits and dangers of the GenAI tools you use or provide, and proactively communicate how you used it.
4. **Diversity, non-discrimination and fairness:** Be aware of bias in GenAI data and output. Make sure output is in line with the UU values regarding Equality, Diversity and Inclusion.
5. **Environmental and societal well-being:** Whenever possible, choose GenAI tools and instruments designed with climate and social considerations in mind. Do not use GenAI if your purpose can be achieved as effectively by other means not involving the use of GenAI.

AI Policy - Utrecht University



Researchers and research assistants



Employees

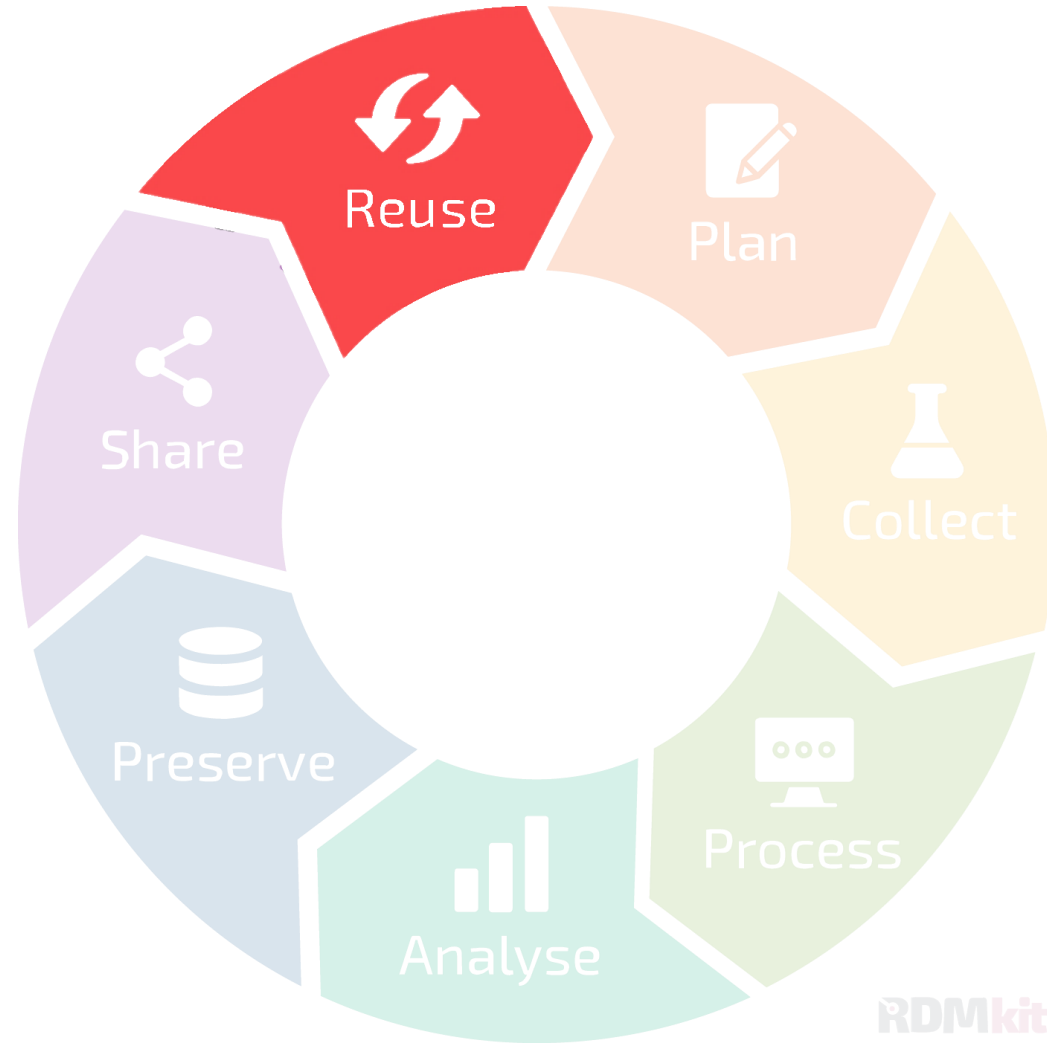


Students



Teachers and educational support staff

Data Reuse



Exploring published datasets



PRIDE
PRoteomics IDentifications Database

[Home](#) [Resources](#) [Tools](#) [Help](#) [License](#) [About](#) [Contact](#) [Log in](#) [Register](#)

Archive [Search](#)

[PXD005011](#)> [cancer](#)> [human](#)> [P02768](#)> [YLEVEPVS](#)>



PRIDE ARCHIVE



PRIDE ARCHIVE USI

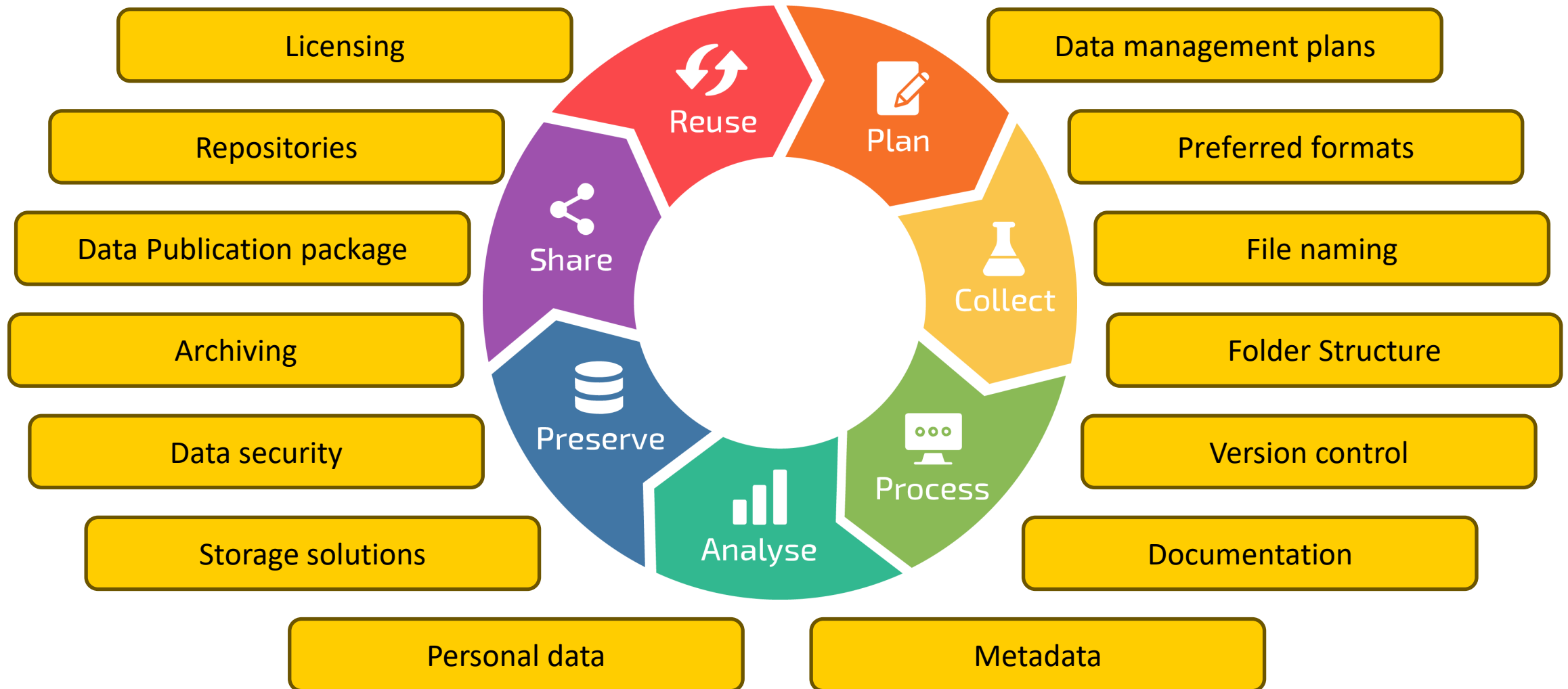


PRIDE CROSSLINKING



[Proteomic and Phosphoproteomic Analyses during Plant Regeneration Initiation in Cotton \(Gossypium hirsutum L.\)](#)

Research Data Management (RDM)



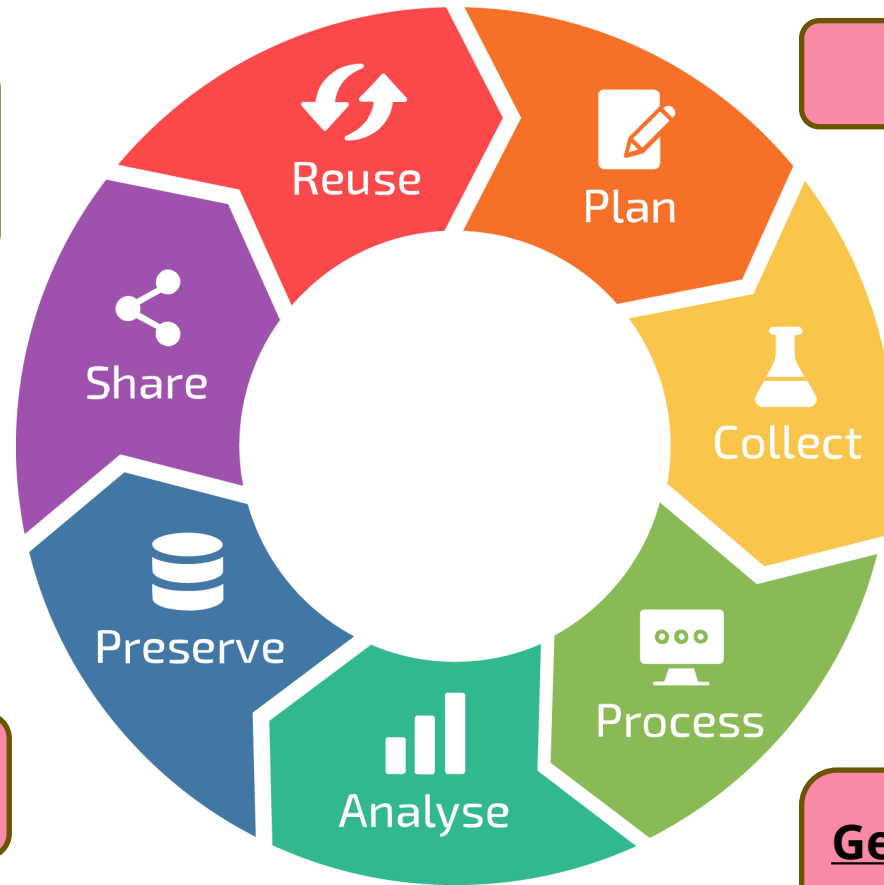
Quickstart to Research Data Management

Writing Reproducible Manuscripts in R and Python

Best Practices in Writing Reproducible Code

Getting started with Yoda

Privacy basics for researchers



Write your DMP

Introduction to Python

Introduction to R & Data

Introduction to GitHub

Getting started with Virtual Research Environments

Got a question? Need to consult an expert?

 **FAIR data and Software**

 **Data publishing** (*YODA, DataverseNL, other repositories*)

 **Data storage** (*YODA, One-drive, Surf-drive, research drive, U/O: drive ...*)

 **Data management plans (DMP)**

 **Metadata**

 **Reproducibility** *of your data & software workflow and other best practices.*

 **High performance computing**

 **Machine Learning**

 **Privacy (GDPR, informed consent)**

 **GitHub**

 **R and Data, Python and Data**

 **...anything data related!**



**Utrecht
University**

Sharing science,
shaping tomorrow

RDM WALK-IN HOURS

Utrecht University Library – Room 1.14

Every Monday | 15:00 – 17:00

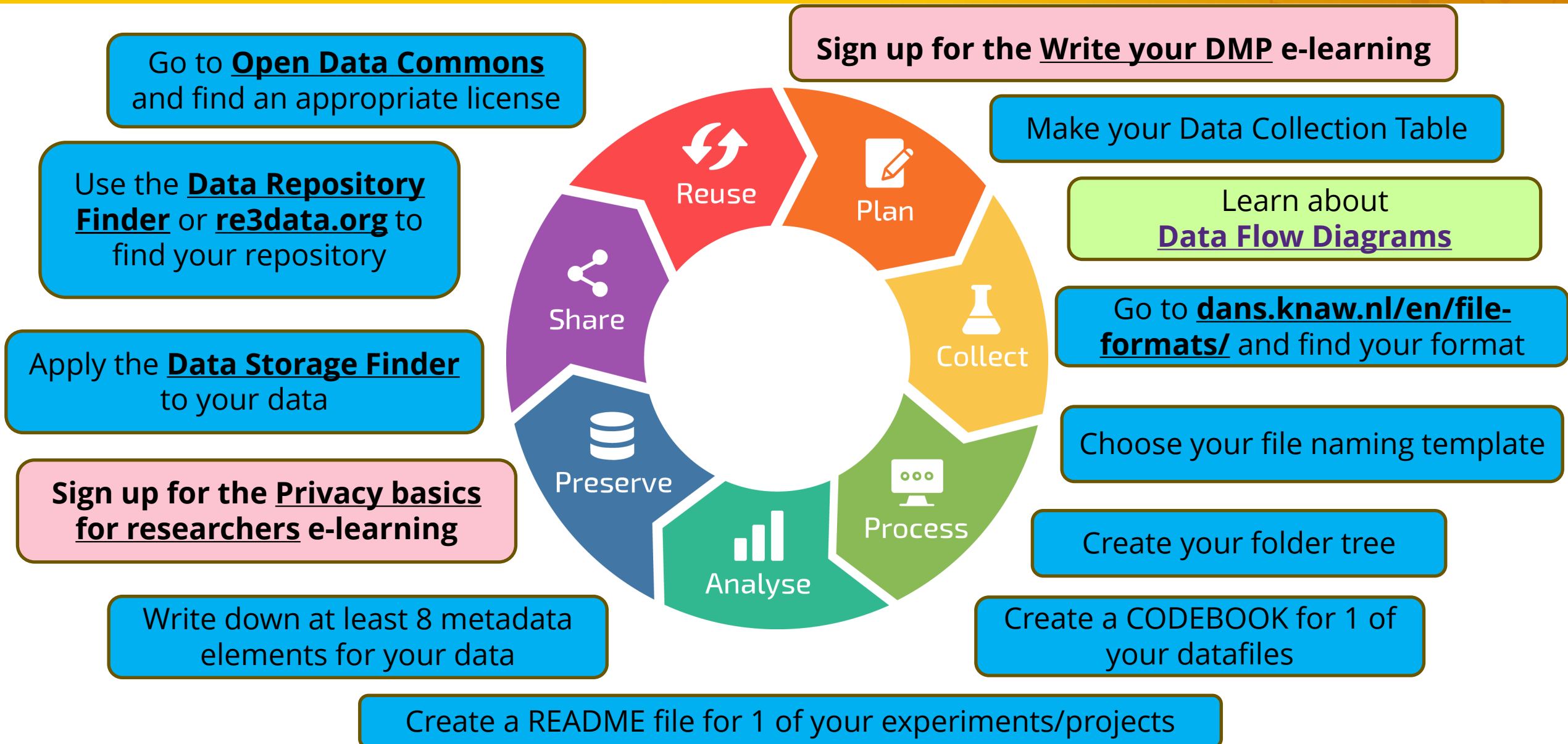
Contact us at: info.rdm@uu.nl

Website: www.uu.nl/rdm



Research Data Management Support

Let's get a Quick Start!





**Utrecht
University**

Sharing science,
shaping tomorrow